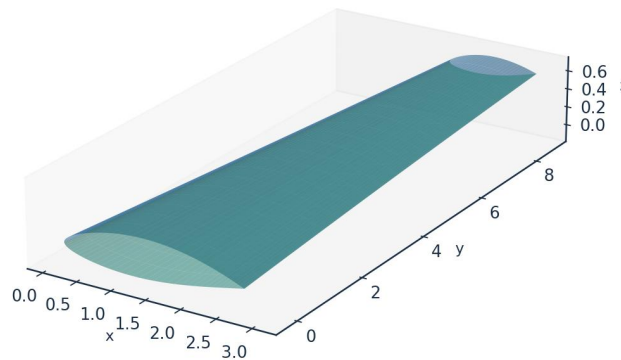


## INDUSTRIAL CASE STUDY

# Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator  
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

*Prepared by*

**AKOSA SAMUEL ONYEJEKWE**



## 1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

**Key results at the cruise design point (FL360,  $M=0.42$ ,  $Re_{MAC} \approx 6.4 \times 10^6$ ):**

| Quantity                                           | Predicted value                       |
|----------------------------------------------------|---------------------------------------|
| Upper-surface transition $x_{tr}/c$                | 0.45 (natural / Tollmien–Schlichting) |
| Lower-surface transition $x_{tr}/c$                | 0.58 (natural / Tollmien–Schlichting) |
| Mean laminar-flow extent                           | $\approx 52\%$ of chord               |
| Section lift coefficient $C_l$                     | 0.502                                 |
| Section profile drag $C_d$                         | 49.6 counts                           |
| Viscous drag reduction vs fully-turbulent          | $\approx 43\%$                        |
| Climb ( $Tu=0.9\%$ ) transition $x_{tr}/c$ (upper) | 0.05 (bypass)                         |

*Table 1. Headline predictions.*

Validated against three independent, credible published transition datasets with one universal calibration set, the solver reproduces the transition-onset Reynolds number  $Re_{\theta t}$  against the Rolls-Royce hot-wire measurements of ERCOFTAC case 020, the Schubauer-Skramstad experiment, and two independent swept-wing experiments. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

## 2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, governed by the growth of the amplification envelope and predicted here by an  $e^N$  integral in  $Re_\theta$ .
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

## 3. Problem Statement and Solution Approach

### 3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction  $C_f$ , momentum thickness  $\theta$ , shape factor  $H$ , intermittency  $\gamma$ ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

### 3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure  $C_p$  and edge velocity  $U_e$ , with a Prandtl–Glauert correction applied to  $C_p$  and the integrated loads at the cruise Mach number. The boundary-layer closures downstream are incompressible formulations and the edge velocity passed to them is left uncorrected.
- Laminar boundary layer: Thwaites' integral method advances  $\theta$ ,  $H$  and  $Re_\theta$  from the stagnation point along each surface.
- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through  $\gamma$ .
- Turbulent boundary layer: Head's entrainment method with the Ludwig–Tillmann skin-friction law advances the turbulent BL to the trailing edge.

- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

## 4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

### 4.1 Inviscid edge solution

(E02) Vortex-panel surface velocity (Kuethe & Chow)

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^N C_{t,ij} \gamma_j$$

(E03) Kutta condition (sharp trailing edge)

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) Pressure coefficient (inviscid)

$$C_p = 1 - \left( \frac{V}{U_\infty} \right)^2$$

[missing eq E04]

### 4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6} \int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu} \frac{dU_e}{dx}, Re_\theta = \frac{U_e \theta}{\nu}$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2 + H) \frac{\theta}{U_e} \frac{dU_e}{dx} = \frac{C_f}{2}$$

### 4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the local  $Tu$ ; natural/TS onset integrates the envelope amplification rate of Drela & Giles in  $Re_\theta$  and triggers at  $N_{crit}$ ; envelope; separation-induced and cross-flow onsets use bubble and C1 criteria. The kernel takes the minimum effective onset across all four:

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp \left[ F(\lambda_\theta) - \frac{F(\lambda_\theta) Tu}{6.91} \right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (low-Tu correlation surrogate)

$$N(x) = \int_{x_0}^x -\alpha_i(x') dx' \geq N_{crit}$$

(E11) Cross-flow criterion (swept wing, C1 on Re\_theta2)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin \Lambda \cos \Lambda \geq C_1$$

(E12) Separation-induced onset (laminar bubble)

$$\lambda \leq -0.09 \Rightarrow Re_{\theta t, sep} = \max(Re_{\theta t}^{floor}, 0.7 Re_{\theta t, bp})$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^* = \min(a_{TS} Re_{\theta t}^{TS}, a_{BP} Re_{\theta t}^{BP}, a_{SEP} Re_{\theta t}^{SEP}, a_{CF} Re_{\theta t}^{CF})$$

(E14) Transition trigger

$$Re_\theta(x_t) \geq Re_{\theta t}^*$$

## 4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412\xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} \sim \frac{\nu}{U_e} C_{len} Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi = (1 - \gamma)\phi_{lam} + \gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e} \frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwig-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

## 4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2 \frac{\theta_{TE}}{c} \left( \frac{U_{e,TE}}{U_\infty} \right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r \frac{\gamma - 1}{2} M_e^2 \left[ 1 - \left( \frac{u}{U_e} \right)^2 \right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e \left( 1 + r \frac{\gamma - 1}{2} M_e^2 \right), r \approx Pr^{1/3}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

## 5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

| Capability                   | XFOIL / e <sup>N</sup> codes | RANS $\gamma$ -Re <sub><math>\theta</math></sub> (CFD) | LES / DNS   | UTSS (this work)               |
|------------------------------|------------------------------|--------------------------------------------------------|-------------|--------------------------------|
| Natural / TS transition      | Yes                          | Indirect                                               | Yes         | Yes (envelope e <sup>N</sup> ) |
| Bypass (free-stream Tu)      | No                           | Yes                                                    | Yes         | Yes (AGS)                      |
| Separation-induced           | Limited                      | Yes                                                    | Yes         | Yes (bubble criterion)         |
| Cross-flow (3-D swept)       | No                           | Add-on only                                            | Yes         | Yes (C1, built-in)             |
| Single universal calibration | n/a                          | Needs re-tuning                                        | n/a         | Yes (one constant set)         |
| 3-D wing capability          | 2-D only                     | Yes                                                    | Yes         | Yes (strip + cross-flow)       |
| Robustness / convergence     | Can fail in sep.             | Stiff, costly                                          | Very costly | Robust, direct                 |
| Typical CPU cost             | seconds                      | hours                                                  | days-weeks  | < 1 second                     |
| Suited to design loops       | Partly                       | No                                                     | No          | Yes                            |

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e<sup>N</sup> codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. The separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here — the novelty claim.

## 6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

## 6.1 Aircraft and wing geometry (input)

| parameter                  | value         | unit           |
|----------------------------|---------------|----------------|
| Aircraft                   | AETHER-NLF 25 | -              |
| Wing reference area S      | 33.150        | m <sup>2</sup> |
| Wing span b                | 17.00         | m              |
| Aspect ratio AR            | 8.72          | -              |
| Root chord c_root          | 2.600         | m              |
| Tip chord c_tip            | 1.300         | m              |
| Taper ratio                | 0.500         | -              |
| Mean aerodynamic chord MAC | 2.022         | m              |
| Leading-edge sweep         | 12.0          | deg            |
| Dihedral                   | 4.0           | deg            |
| Tip washout (twist)        | -3.0          | deg            |
| Section                    | UTSS-NLF16    | -              |
| Section max thickness      | 16.0          | % chord        |
| Max-thickness location     | 42.2          | % chord        |
| Section design Cl          | 0.45          | -              |

Table 3. Geometry definition (input).

## 6.2 Flight conditions (input)

| case                     | altitude_m | mac_h | U_inf_ms | rho_kgm3 | mu_Pas    | T_inf_K | Tu_pct | alpha_deg | Re_MAC     | nu_m2s    | q_inf_Pa |
|--------------------------|------------|-------|----------|----------|-----------|---------|--------|-----------|------------|-----------|----------|
| CRUISE<br>(FL360, M0.42) | 11000.0    | 0.42  | 123.93   | 0.3639   | 1.422e-05 | 216.65  | 0.07   | 1.5       | 6414000.0  | 3.907e-05 | 2794.6   |
| CLIMB<br>(FL100, M0.30)  | 3000.0     | 0.3   | 98.57    | 0.9093   | 1.694e-05 | 268.66  | 0.9    | 3.5       | 10700000.0 | 1.863e-05 | 4417.8   |

Table 4. Flight / flow conditions (input).

Flight-condition comparison: cruise vs climb (flow\_conditions.csv)

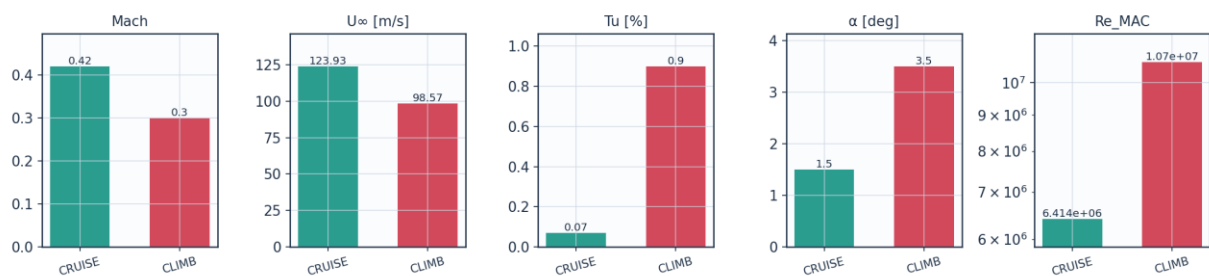


Fig. 8a. Cruise vs climb flight-condition comparison (flow\_conditions.csv).

## 6.3 Fluid (material) properties (input)

| property                | value           | unit   |
|-------------------------|-----------------|--------|
| Working fluid           | air (ideal gas) | -      |
| Specific gas constant R | 287.05          | J/kg.K |

|                               |          |        |
|-------------------------------|----------|--------|
| Ratio of specific heats gamma | 1.40     | -      |
| Prandtl number Pr             | 0.72     | -      |
| Sutherland reference mu0      | 1.716e-5 | Pa.s   |
| Sutherland reference T0       | 273.15   | K      |
| Sutherland constant S         | 110.4    | K      |
| Recovery factor (turbulent)   | 0.89     | -      |
| Cruise dynamic viscosity      | 1.422e-5 | Pa.s   |
| Cruise density                | 0.36392  | kg/m^3 |

Table 5. Air properties (input).

## 6.4 Solver settings (input)

| setting                 | value                                                                  |
|-------------------------|------------------------------------------------------------------------|
| Inviscid method         | Constant-strength vortex-panel (Kuethe-Chow)                           |
| Kutta condition         | Enforced at sharp trailing edge                                        |
| Surface panels          | 260 (cosine-clustered)                                                 |
| Compressibility         | none - incompressible panel solution                                   |
| Laminar BL closure      | Thwaites integral method                                               |
| Transition model        | UTSS unified 4-mechanism kernel                                        |
| mechanisms              | TS/natural(AGS@Tu=0.03%)   bypass(AGS@Tu)   separation   crossflow(C1) |
| Intermittency closure   | Narasimha universal (gamma)                                            |
| Turbulent BL closure    | Head entrainment + Ludwig-Tillmann Cf                                  |
| Separation criterion    | Thwaites lambda<=-0.09 / H>2.6 turbulent                               |
| Drag integration        | Squire-Young far-wake                                                  |
| Marching scheme         | explicit, arc-length stepping                                          |
| Convergence tol (panel) | 1e-10 (direct solve)                                                   |
| Wall y+ target          | ~1 (reconstruction grid)                                               |

Table 6. Solver configuration (input).

## 6.5 Universal calibration constants (input)

| constant  | value   | role                                 | calibration_source                                    |
|-----------|---------|--------------------------------------|-------------------------------------------------------|
| A_TS      | 1.0     | TS/natural onset weight              | validated on Schubauer-Skramstad (NACA Rep.909)       |
| A_BP      | 1.0     | bypass onset weight                  | validated on ERCOFTAC T3A/T3B                         |
| A_SEP     | 1.0     | separation-induced weight            | Mayle (1991) bubble correlation                       |
| A_CF      | 1.0     | crossflow weight                     | Arnal C1 criterion                                    |
| N_crit    | from Tu | e^N amplification factor             | Mack correlation N=-8.43-2.4 ln(Tu); 9.00 at Tu=0.07% |
| N_floor   | 0.5     | lower clamp on N_crit at high Tu     | clamp; bypass governs there                           |
| C_len     | 6.5     | Narasimha transition-length scale    | Dhawan-Narasimha (1958)                               |
| CF_C1     | 150.0   | crossflow critical Re_theta2         | Arnal et al. C1 criterion                             |
| CF_ratio  | 0.47    | theta2/theta surrogate for crossflow | calibrated, not validated                             |
| sep_floor | 120.0   | min separation onset Re_theta        | calibration floor                                     |

Table 7. Calibration constants and their sources (input).

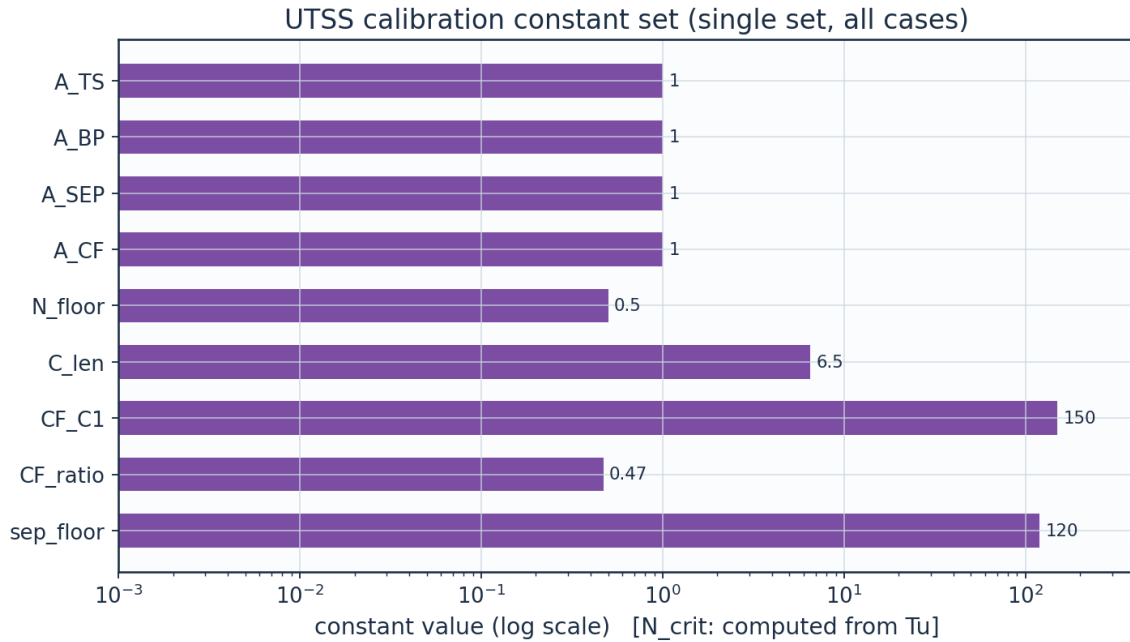


Fig. 8b. UTSS universal calibration constant set (calibration\_constants.csv).

## 7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

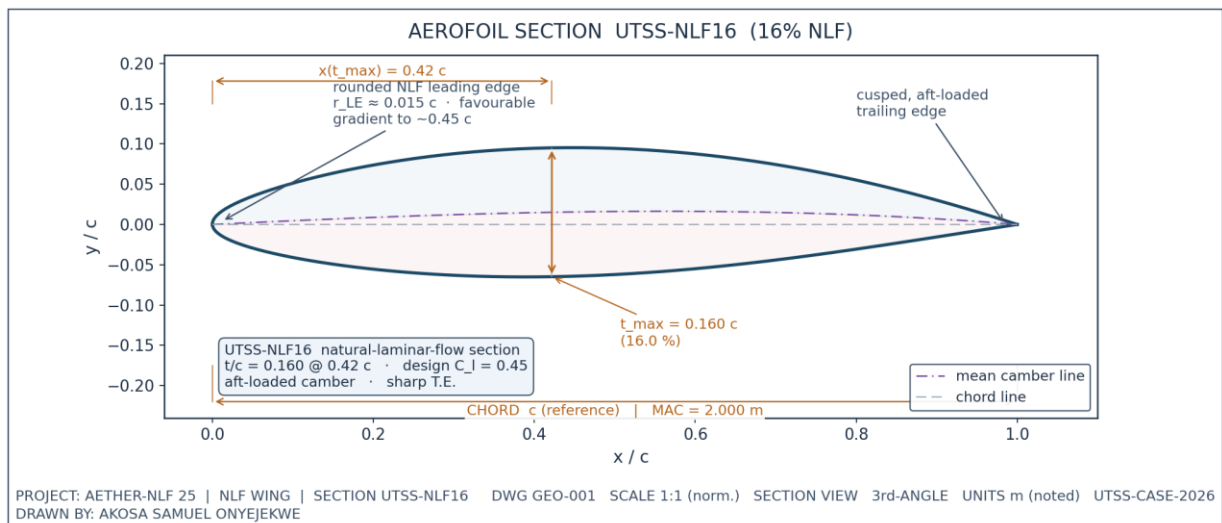
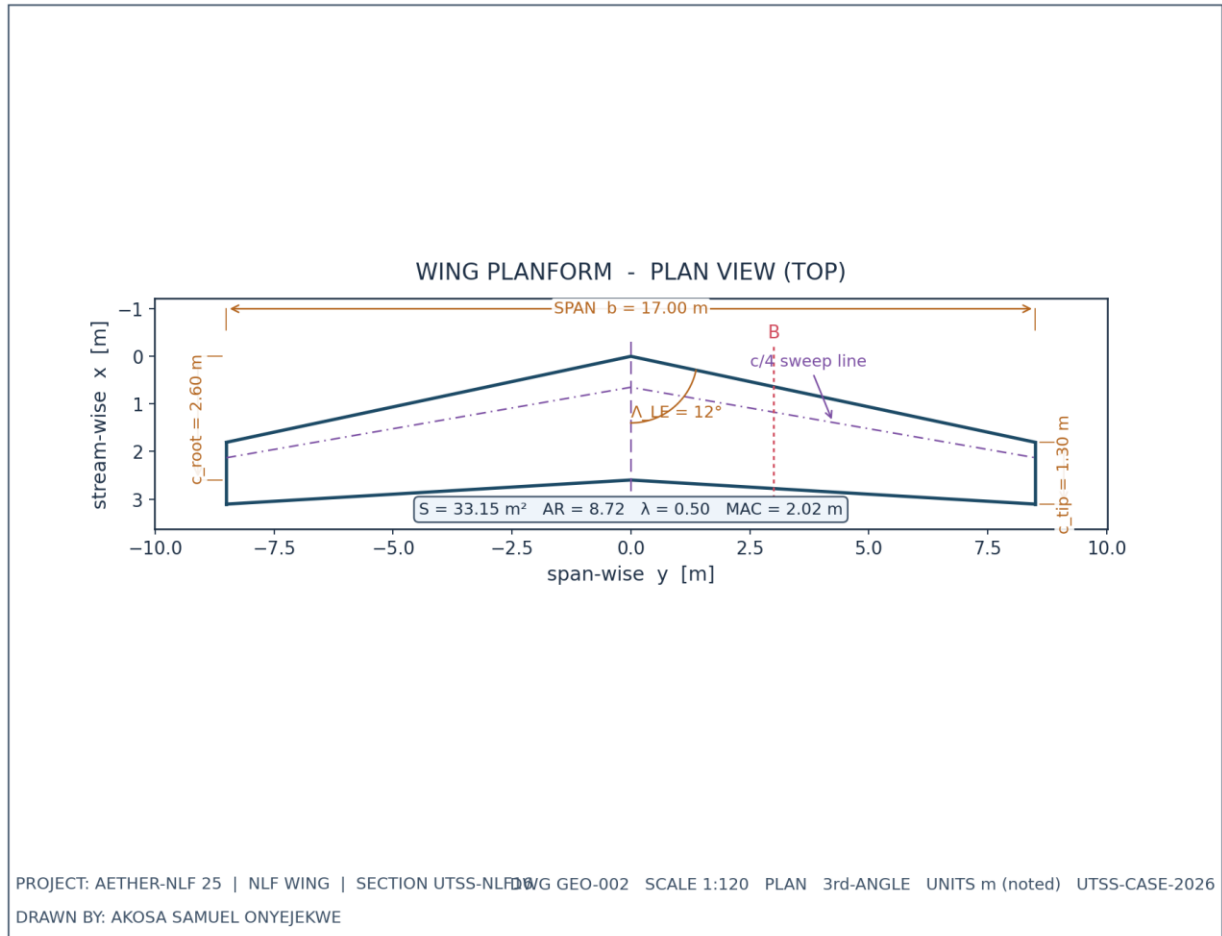
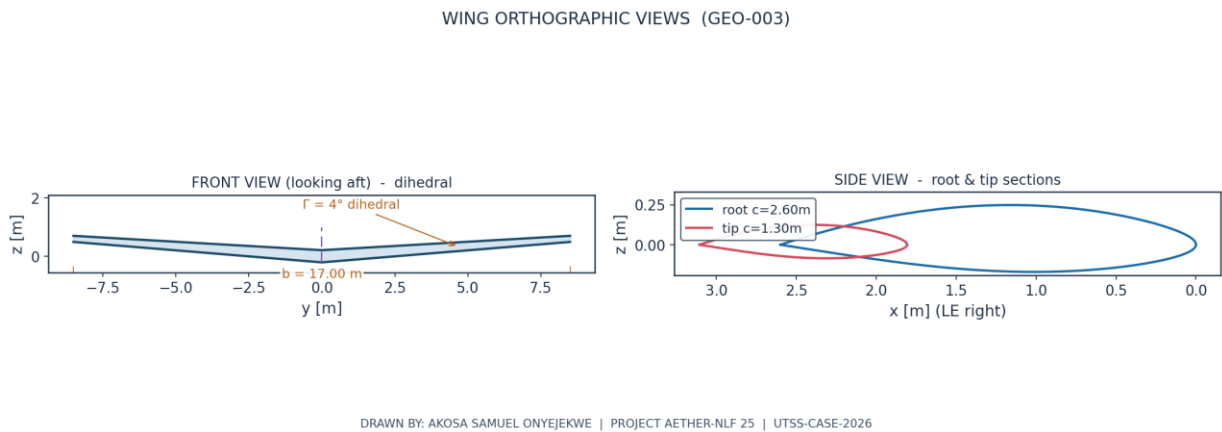


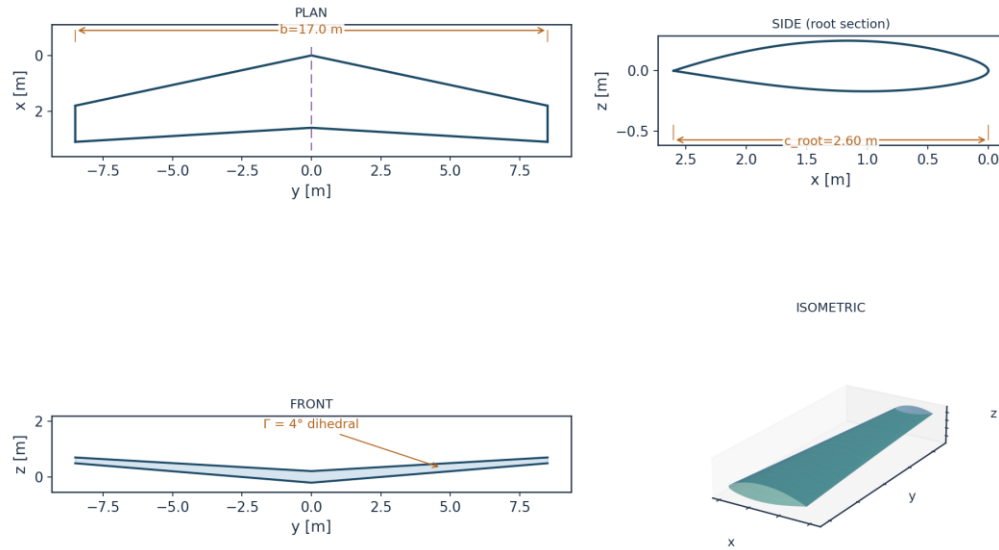
Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).



*Fig. 2. Wing planform — plan (top) view, dimensioned.*



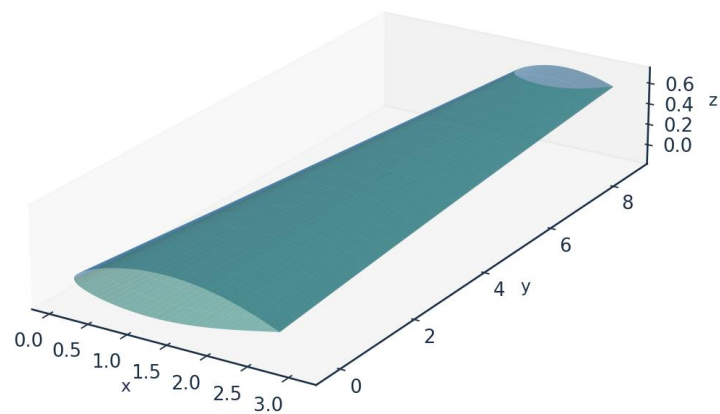
*Fig. 3. Front and side orthographic views.*



All dimensions in metres unless noted | Scale 1:120 | UTSS-CASE-2026 | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

*Fig. 4. Full third-angle orthographic projection sheet.*

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 5. Isometric view of the wing.

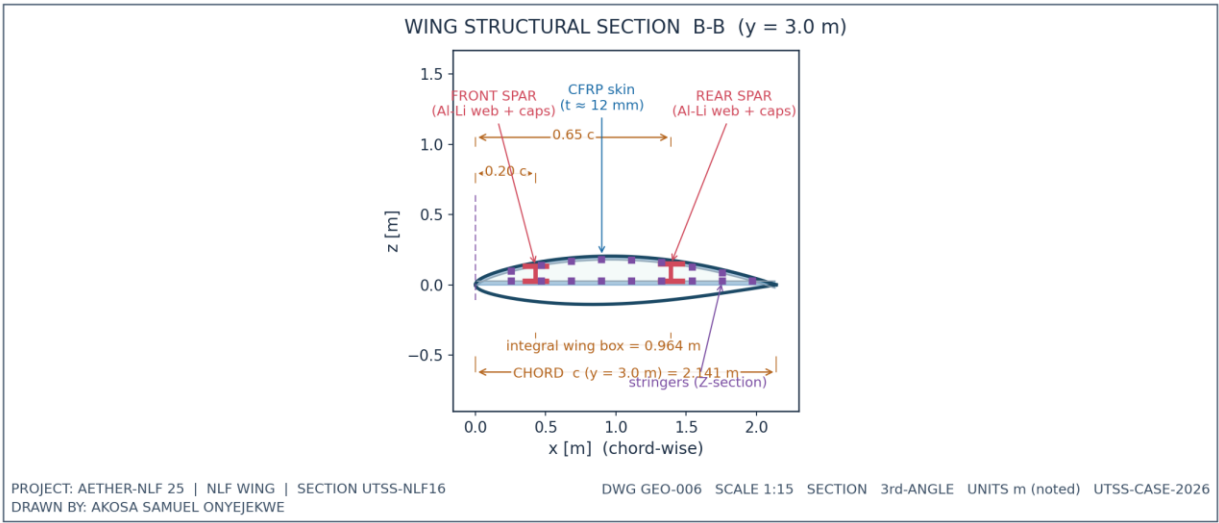


Fig. 6. Structural sectional view B-B.

## 7.1 Geometry data (from CSV)

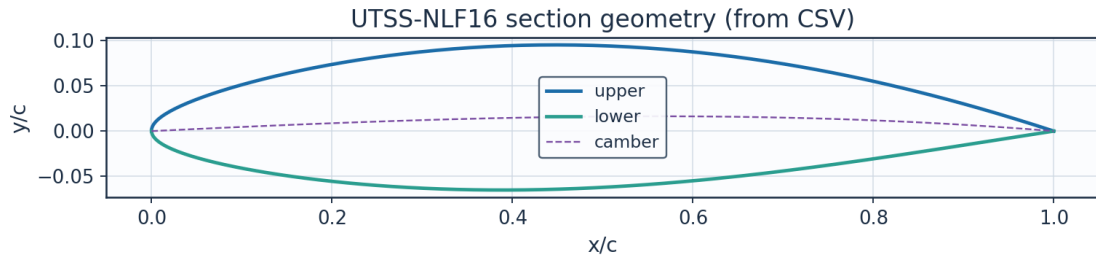


Fig. 7. Section geometry plotted from airfoil\_UTSS-NLF16.csv.

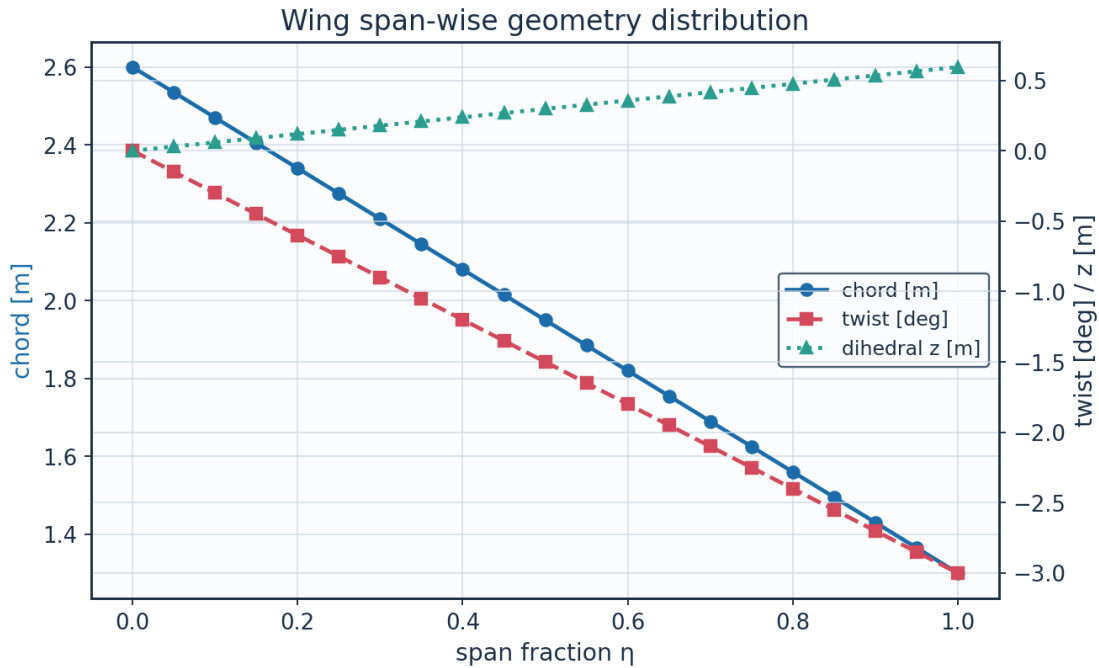


Fig. 8. Span-wise geometry from wing\_planform.csv.

| eta  | y_m                | chord_m | x_le_m             | x_te_m             | twist_deg           | z_dihedral_m       | Re_local           |
|------|--------------------|---------|--------------------|--------------------|---------------------|--------------------|--------------------|
| 0.0  | 0.0                | 2.6     | 0.0                | 2.6                | -0.0                | 0.0                | 8246203.012995781  |
| 0.05 | 0.425              | 2.535   | 0.0903365387097594 | 2.62533653870976   | -0.15               | 0.0297188950759919 | 8040047.9376708865 |
| 0.1  | 0.8500000000000001 | 2.47    | 0.1806730774195188 | 2.650673077419519  | -0.3                | 0.0594377901519838 | 7833892.862345993  |
| 0.15 | 1.275              | 2.405   | 0.2710096161292782 | 2.6760096161292783 | -0.45               | 0.0891566852279757 | 7627737.787021098  |
| 0.2  | 1.7000000000000002 | 2.34    | 0.3613461548390376 | 2.7013461548390376 | -0.6000000000000001 | 0.1188755803039677 | 7421582.711696202  |
| 0.25 | 2.125              | 2.275   | 0.451682693548797  | 2.726682693548797  | -0.75               | 0.1485944753799596 | 7215427.636371308  |
| 0.3  | 2.5500000000000003 | 2.21    | 0.5420192322585565 | 2.7520192322585566 | -0.9000000000000001 | 0.1783133704559515 | 7009272.561046414  |

|                                |                        |       |                        |                        |                             |                        |                        |
|--------------------------------|------------------------|-------|------------------------|------------------------|-----------------------------|------------------------|------------------------|
| <b>0.35</b>                    | 2.975                  | 2.145 | 0.632355770<br>9683159 | 2.777355770<br>968316  | -1.05                       | 0.208032265<br>5319434 | 6803117.485<br>721519  |
| <b>0.4</b>                     | 3.400000000<br>0000004 | 2.08  | 0.722692309<br>6780753 | 2.802692309<br>678076  | -<br>1.200000000<br>0000002 | 0.237751160<br>6079354 | 6596962.410<br>396625  |
| <b>0.45</b>                    | 3.825                  | 2.015 | 0.813028848<br>3878346 | 2.828028848<br>387835  | -1.35                       | 0.267470055<br>6839273 | 6390807.335<br>07173   |
| <b>0.5</b>                     | 4.25                   | 1.95  | 0.903365387<br>097594  | 2.853365387<br>097594  | -1.5                        | 0.297188950<br>7599192 | 6184652.259<br>7468365 |
| <b>0.55</b>                    | 4.675000000<br>000001  | 1.885 | 0.993701925<br>8073536 | 2.878701925<br>807354  | -1.65                       | 0.326907845<br>8359112 | 5978497.184<br>421942  |
| <b>0.600000000<br/>0000001</b> | 5.100000000<br>0000005 | 1.82  | 1.084038464<br>517113  | 2.904038464<br>5171127 | -<br>1.800000000<br>0000005 | 0.356626740<br>9119031 | 5772342.109<br>097046  |
| <b>0.65</b>                    | 5.525                  | 1.755 | 1.174375003<br>2268725 | 2.929375003<br>2268724 | -1.95                       | 0.386345635<br>987895  | 5566187.033<br>772152  |
| <b>0.700000000<br/>0000001</b> | 5.95                   | 1.69  | 1.264711541<br>9366315 | 2.954711541<br>936632  | -2.1                        | 0.416064531<br>0638869 | 5360031.958<br>447257  |
| <b>0.75</b>                    | 6.375                  | 1.625 | 1.355048080<br>6463912 | 2.980048080<br>646391  | -2.25                       | 0.445783426<br>1398788 | 5153876.883<br>122363  |
| <b>0.8</b>                     | 6.800000000<br>000001  | 1.56  | 1.445384619<br>3561506 | 3.005384619<br>35615   | -<br>2.400000000<br>0000004 | 0.475502321<br>2158709 | 4947721.807<br>797468  |
| <b>0.850000000<br/>0000001</b> | 7.225                  | 1.495 | 1.535721158<br>06591   | 3.030721158<br>06591   | -<br>2.550000000<br>0000003 | 0.505221216<br>2918628 | 4741566.732<br>472573  |
| <b>0.9</b>                     | 7.65                   | 1.43  | 1.626057696<br>7756692 | 3.056057696<br>775669  | -2.7                        | 0.534940111<br>3678547 | 4535411.657<br>14768   |
| <b>0.95</b>                    | 8.075000000<br>000001  | 1.365 | 1.716394235<br>485429  | 3.081394235<br>485429  | -2.85                       | 0.564659006<br>4438467 | 4329256.581<br>822785  |
| <b>1.0</b>                     | 8.5                    | 1.3   | 1.806730774<br>195188  | 3.106730774<br>195188  | -3.0                        | 0.594377901<br>5198385 | 4123101.506<br>49789   |

Table 8. Wing planform stations (wing\_planform.csv).

## Lofted wing sections (from wing\_sections\_3d.csv)

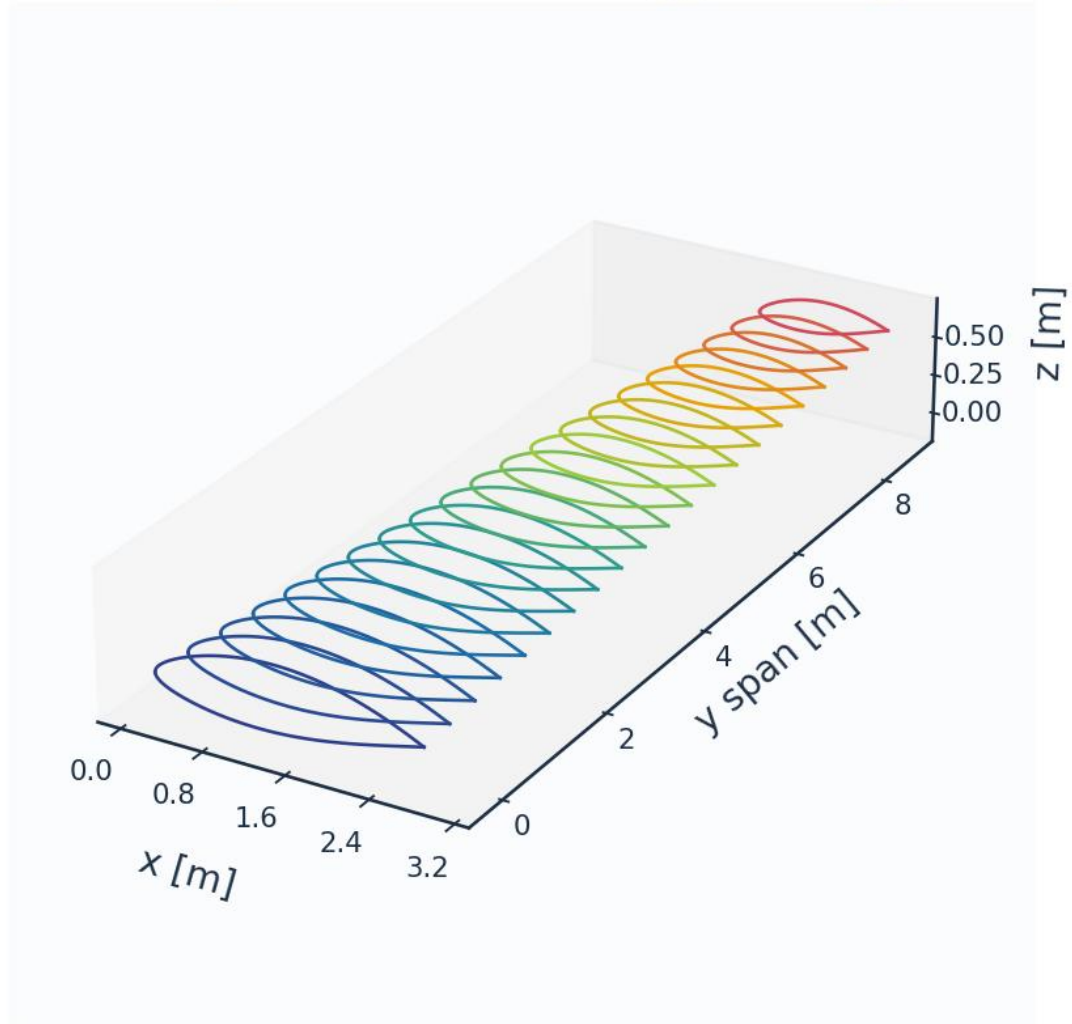


Fig. 8c. Lofted 3-D wing sections (wing\_sections\_3d.csv).

## 8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell  $y^+ \approx 1$ ) supports boundary-layer profile recovery. A mesh-independence study confirms convergence.

| metric                              | value | unit    |
|-------------------------------------|-------|---------|
| Surface streamwise nodes            | 321   | -       |
| Wall-normal layers (reconstruction) | 44    | -       |
| First-cell height $y_1$             | 6.51  | micron  |
| Target wall $y^+$                   | 0.80  | -       |
| Wall-normal growth ratio            | 1.12  | -       |
| Min streamwise spacing              | 0.099 | mm (xc) |
| Max streamwise spacing              | 9.927 | mm (xc) |

|                                   |                                                     |     |
|-----------------------------------|-----------------------------------------------------|-----|
| LE clustering ratio               | 99.9                                                | -   |
| BL grid outer extent              | 7.04                                                | mm  |
| Friction velocity $u_{\tau}$ (TE) | 4.800                                               | m/s |
| Max cell aspect ratio             | 1524                                                | -   |
| Grid orthogonality (min)          | 88.5                                                | deg |
| Mesh type                         | structured C-type (surface) + normal reconstruction | -   |

Table 9. Mesh metrics.

| n_surface_panels | Cl                 | Cd                 | x_tr_upper_c       | dCd_pct            |
|------------------|--------------------|--------------------|--------------------|--------------------|
| 120              | 0.4996087807380917 | 0.0044861077973023 | 0.4861367093300009 |                    |
| 180              | 0.5010076226503911 | 0.0048183870669946 | 0.4553399055769377 | 7.40684987311746   |
| 260              | 0.5018338859371162 | 0.0049596252583119 | 0.4445436762963268 | 2.931233820642909  |
| 360              | 0.5023435792157286 | 0.0050600731041123 | 0.442184134913257  | 2.0253111993098427 |
| 480              | 0.5026750206350319 | 0.0051339915261796 | 0.4367159348591741 | 1.460817275688342  |

Table 10. Mesh-independence study.

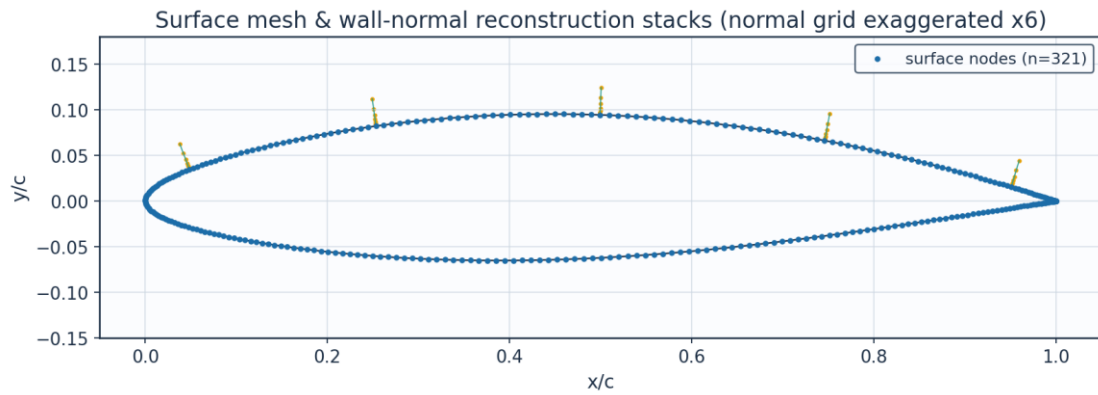


Fig. 9. Surface mesh and wall-normal stacks.

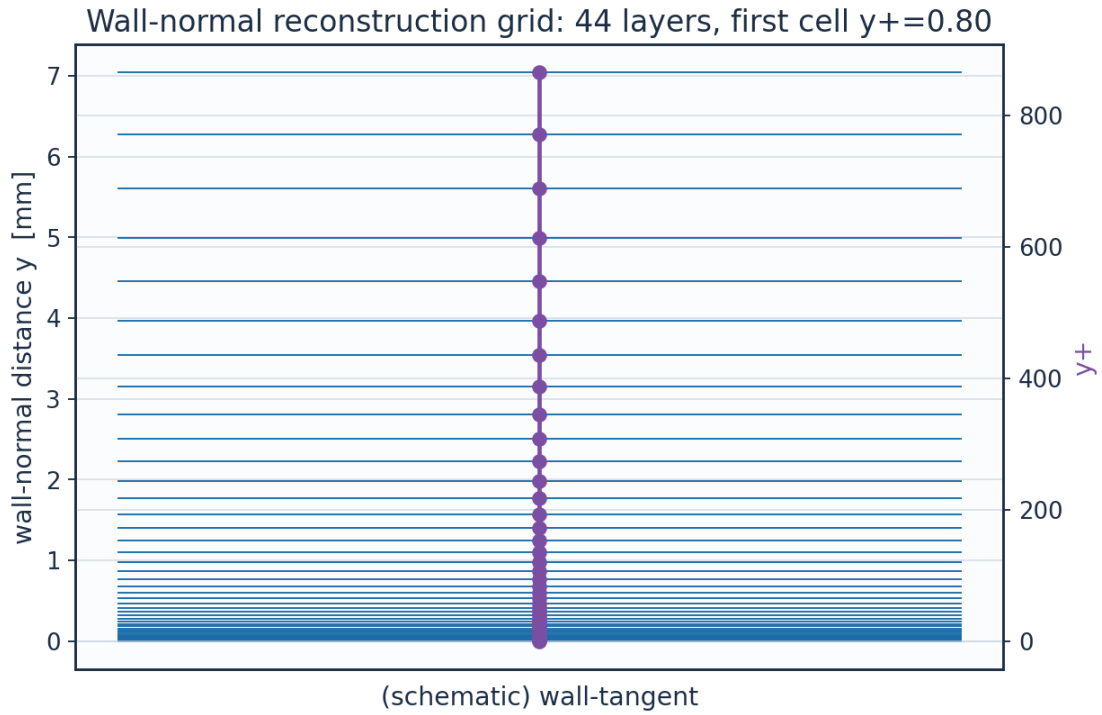


Fig. 10. Wall-normal reconstruction grid /  $y^+$ .

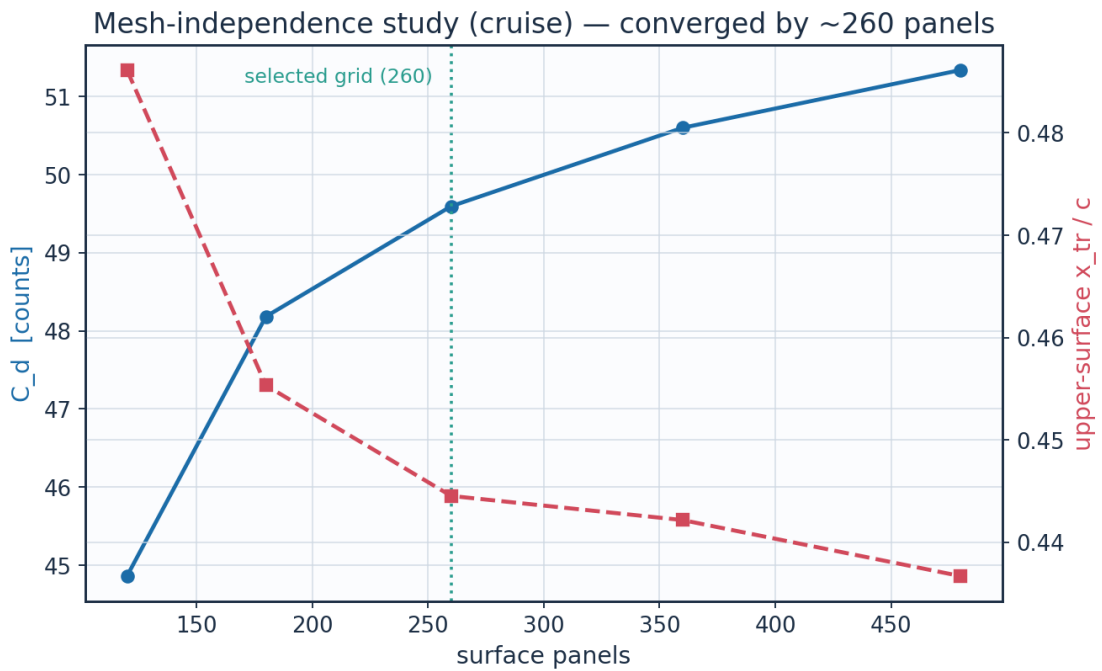


Fig. 11. Mesh-independence convergence.

| layer | $y_m$               | $y_{plus}$ | cell_dy_m              | growth_ratio |
|-------|---------------------|------------|------------------------|--------------|
| 0.0   | 0.0                 | 0.0        | 6.5127391337287455e-06 | 1.0          |
| 2.0   | 1.380700696350494e- | 1.696      | 7.731923899562769e-    | 1.12         |

|      |                        |                    |                        |                    |
|------|------------------------|--------------------|------------------------|--------------------|
|      | 05                     |                    | 06                     |                    |
| 4.0  | 3.112651649852554e-05  | 3.8234624          | 9.698925339611536e-06  | 1.1199999999999997 |
| 6.0  | 5.2852109259255385e-05 | 6.4921512345600005 | 1.2166331946008715e-05 | 1.12               |
| 8.0  | 8.010469281831492e-05  | 9.839754508632067  | 1.526144679307334e-05  | 1.1199999999999997 |
| 10.0 | 0.0001142903336347     | 14.038988055628067 | 1.9143958857231197e-05 | 1.1199999999999997 |
| 12.0 | 0.0001571728014749     | 19.30650661697985  | 2.401418199051081e-05  | 1.12               |
| 14.0 | 0.0002109645691337     | 25.91408190033953  | 3.012338988889677e-05  | 1.12               |
| 16.0 | 0.0002784409624848     | 34.20262433578592  | 3.778678027663213e-05  | 1.1200000000000001 |
| 18.0 | 0.0003630833503045     | 44.599771966809854 | 4.739973717900734e-05  | 1.1199999999999997 |
| 20.0 | 0.0004692587615855     | 57.64195395516631  | 5.945823031734682e-05  | 1.12               |
| 22.0 | 0.0006024451974963     | 74.00206704136062  | 7.458440411007986e-05  | 1.12               |
| 24.0 | 0.0007695142627029     | 94.52419289668278  | 9.355867651568416e-05  | 1.12               |
| 26.0 | 0.000979085698098      | 120.26714756959888 | 0.0001173600038212     | 1.1200000000000006 |
| 28.0 | 0.0012419721066577     | 152.55910991130486 | 0.0001472163887934     | 1.1200000000000008 |
| 30.0 | 0.0015717368175549     | 193.0661474727408  | 0.0001846682381024     | 1.1199999999999997 |
| 32.0 | 0.0019853936709044     | 243.87817538980616 | 0.0002316478378757     | 1.1200000000000003 |
| 34.0 | 0.002504284827746      | 307.6167832089729  | 0.0002905790478312     | 1.1199999999999999 |
| 36.0 | 0.0031551818948881     | 387.57049285733575 | 0.0003645023575995     | 1.12               |
| 38.0 | 0.0039716671759111     | 487.864426240242   | 0.0004572317573729     | 1.12               |
| 40.0 | 0.0049958663124264     | 613.6731362757596  | 0.0005735515164485     | 1.1200000000000014 |
| 42.0 | 0.0062806217092712     | 771.4875821443129  | 0.000719463022233      | 1.1199999999999997 |

Table 11. Wall-normal grid (bl\_normal\_grid.csv, sampled).

(table sampled to 22 rows; full data in 02\_mesh/bl\_normal\_grid.csv)

## 9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

### 9.1 Transition prediction summary

| case   | surface | s_tr_c | x_tr_c | Re_x_tr   | Re_theta_at_onset | mechanism  | laminar_run_pct | Cf_te   |
|--------|---------|--------|--------|-----------|-------------------|------------|-----------------|---------|
| CRUISE | upper   | 0.469  | 0.445  | 3008000.0 | 1234.0            | TS-natural | 44.5            | 0.0002  |
| CRUISE | lower   | 0.587  | 0.578  | 3766000.0 | 1277.1            | TS-natural | 57.8            | 0.00025 |
| CLIMB  | upper   | 0.047  | 0.025  | 506900.0  | 511.4             | bypass     | 2.5             | 0.00014 |
| CLIMB  | lower   | 0.159  | 0.157  | 1706000.0 | 603.7             | bypass     | 15.7            | 0.00019 |

Table 12. Transition prediction summary (transition\_summary.csv).

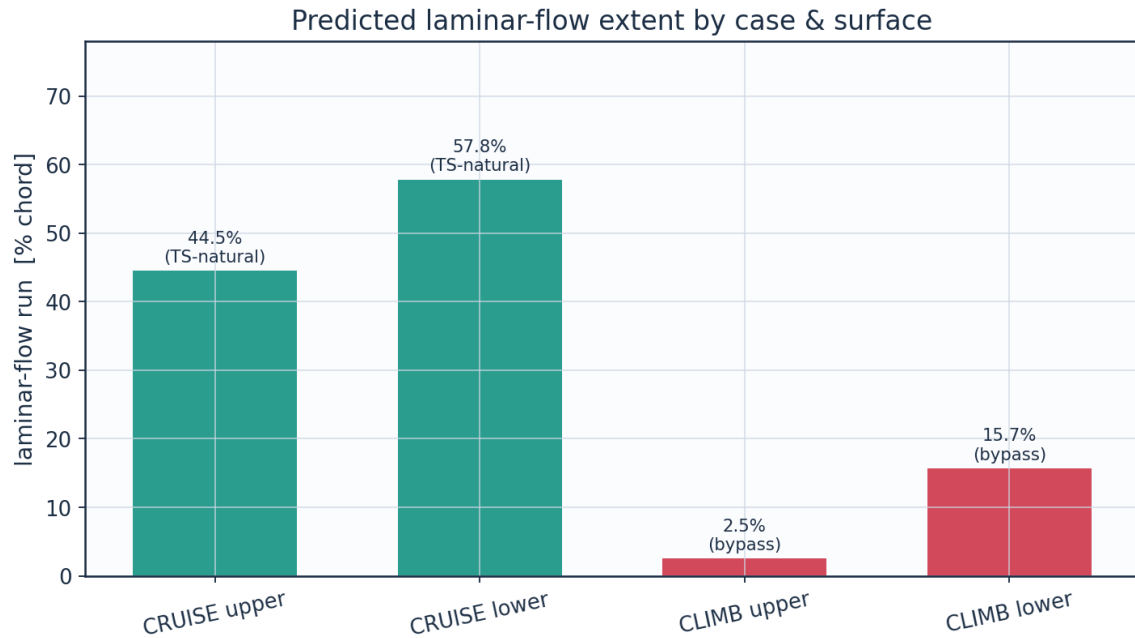


Fig. 12a. Predicted laminar-flow extent by case and surface (transition\_summary.csv).

## 9.2 Integrated forces and drag breakdown

| quantity                    | value     | unit   |
|-----------------------------|-----------|--------|
| Section lift coefficient Cl | 0.5018    | -      |
| Section profile drag Cd     | 0.00496   | -      |
| Section Cd (counts)         | 49.6      | counts |
| Section L/D                 | 101.2     | -      |
| Wing lift (3D est, e=0.85)  | 41842.0   | N      |
| Dynamic pressure q          | 2794.6    | Pa     |
| Reynolds number Re_MAC      | 6414000.0 | -      |
| Mach number                 | 0.42      | -      |

Table 13. Integrated forces.

| configuration                   | Cd_counts | mean_laminar_pct |
|---------------------------------|-----------|------------------|
| NLF (UTSS predicted transition) | 49.6      | 52.8             |
| Fully turbulent (LE trip)       | 87.7      | 0.0              |
| Viscous drag reduction          | 43.4      |                  |

Table 14. NLF vs fully-turbulent drag.

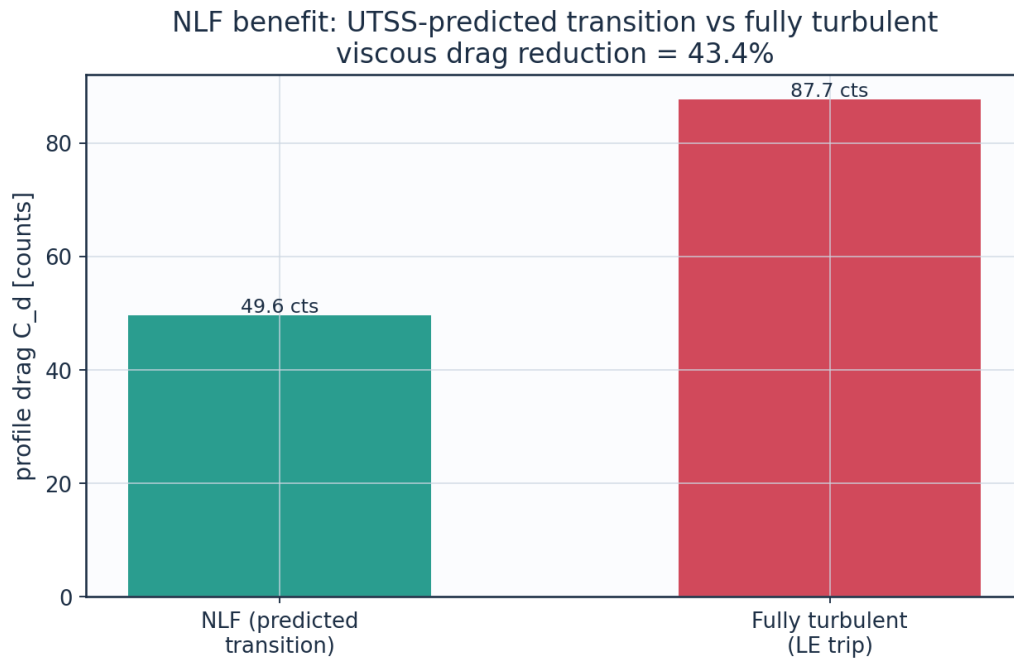


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

### 9.3 Surface distributions — cruise

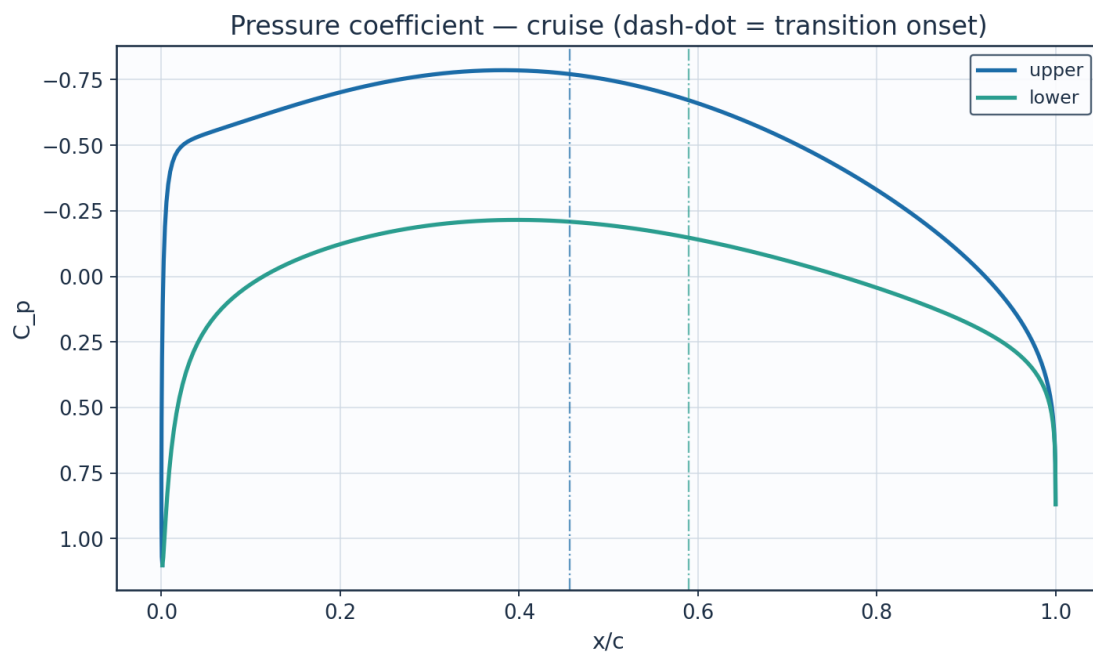


Fig. 13. Pressure coefficient  $C_p$  — cruise.

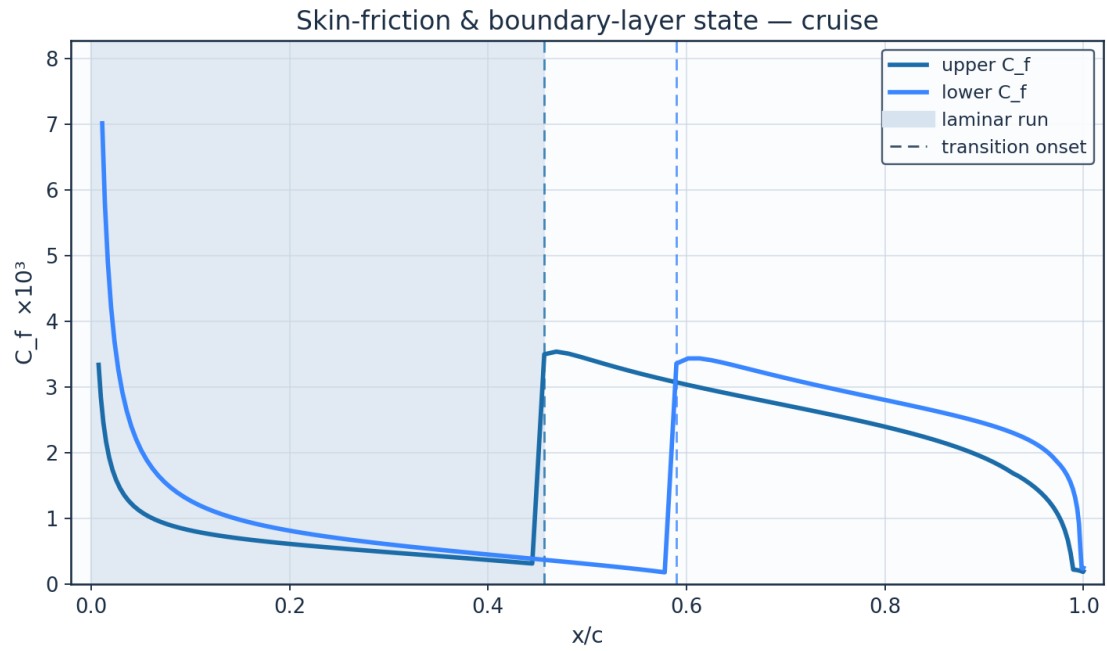


Fig. 14. Skin-friction  $C_f$  and laminar run — cruise.

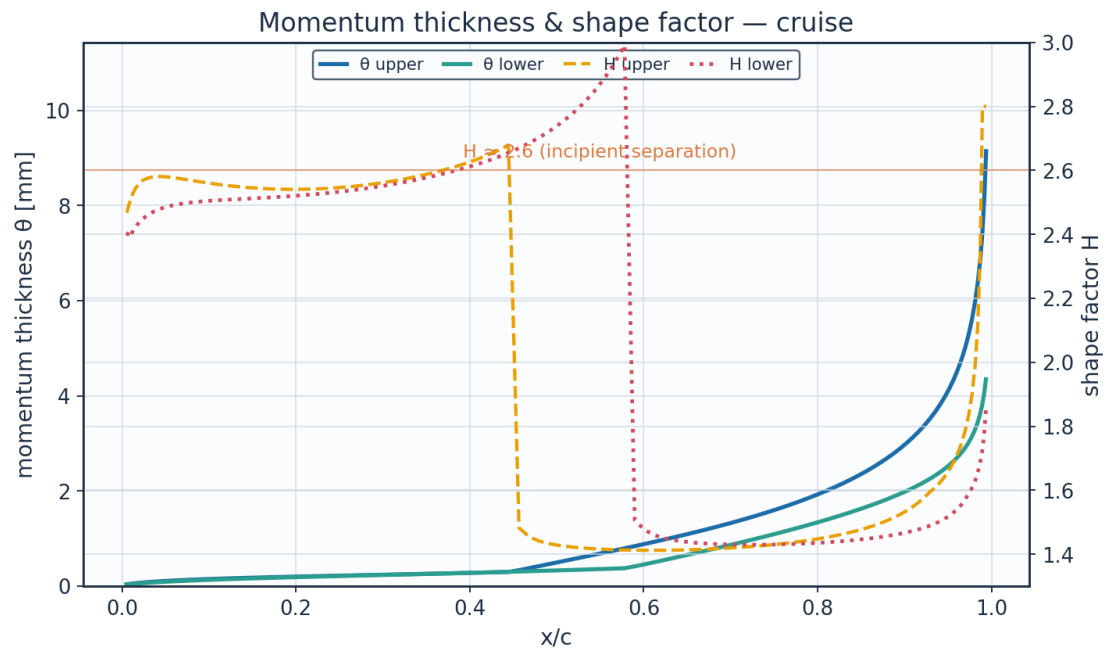


Fig. 15. Momentum thickness  $\theta$  and shape factor  $H$  — cruise.

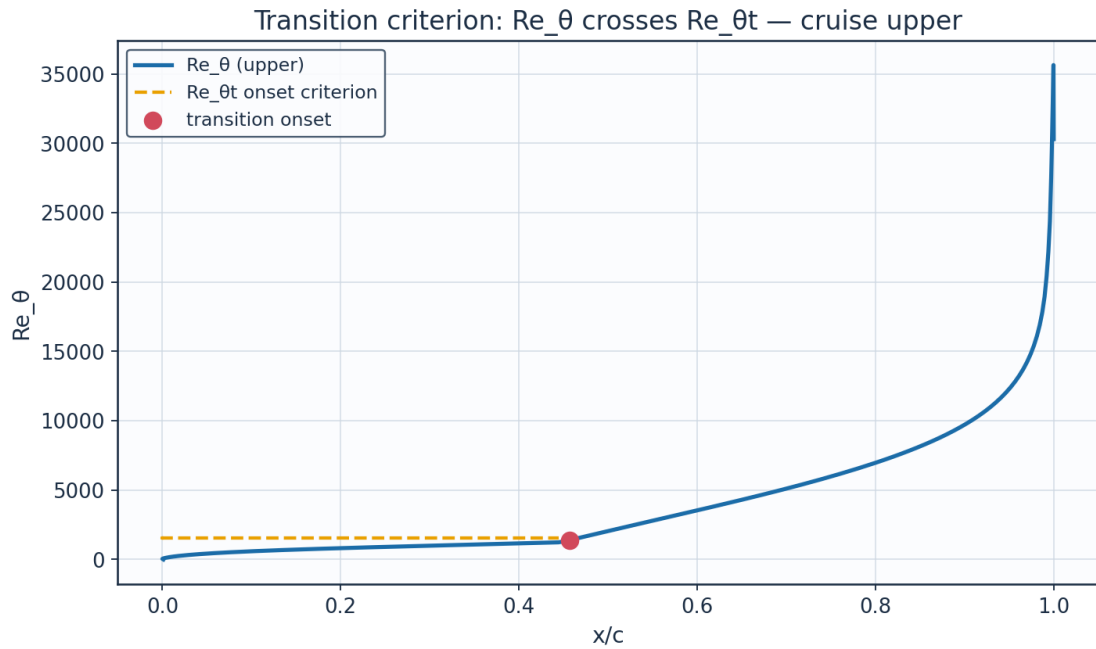


Fig. 16. Transition criterion  $Re_\theta$  vs  $Re_{\theta t}$  — cruise.

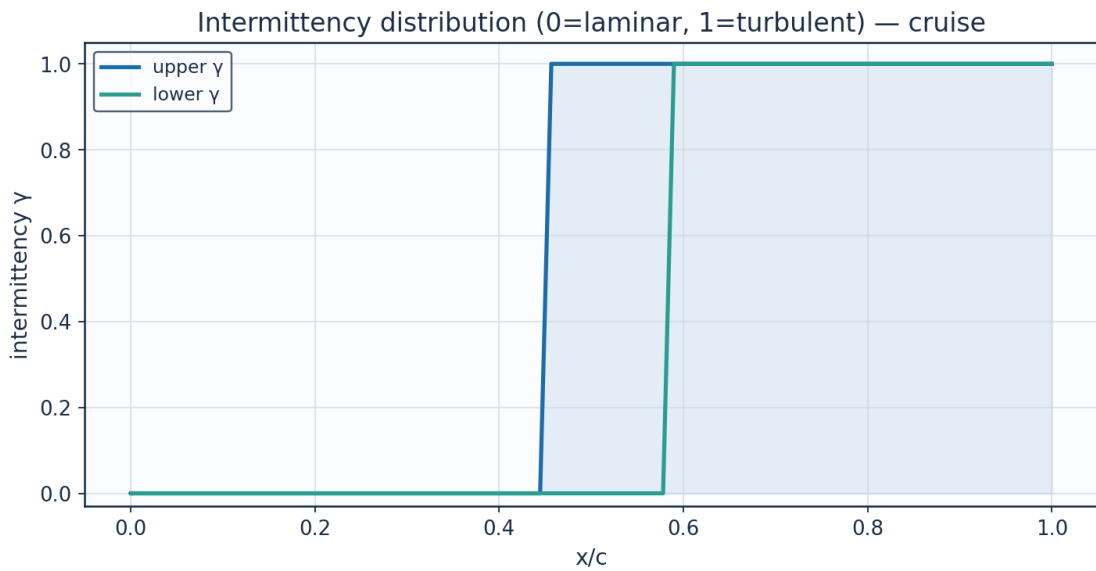


Fig. 17. Intermittency  $\gamma$  — cruise.

| x_c                | arc_s_m     | Re_x        | Cp                 | Ue_ms              | Ue_Uinf            | theta_mm    | H_shape            | Cf                 | Re_theta           | Re_theta_trans    | intermittency_gamma | state   |
|--------------------|-------------|-------------|--------------------|--------------------|--------------------|-------------|--------------------|--------------------|--------------------|-------------------|---------------------|---------|
| 0.0011523200325028 | 0.0         | 0.0         | 1.1018664780939074 | 0.6743779759376758 | 0.0054416302825453 | 1e-05       | 2.6099999414506483 | 2549.4295711287004 | 0.0001725876462751 | 1569.314895047386 | 0.0                 | laminar |
| 0.000242727        | 0.014097598 | 44712.17719 | 0.545321836        | 88.07774049        | 0.710709004        | 0.023652561 | 2.406649053        | 0.011498126        | 53.31515396        | 1569.314895       | 0.0                 | lami    |

|                           |                    |                    |                     |                    |                    |                    |                    |                    |                    |                    |     |                     |
|---------------------------|--------------------|--------------------|---------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-----|---------------------|
| <b>1046013</b>            | 6799438            | 636864             | 2263359             | 458249             | 4378694            | 2246563            | 519997             | 7313345            | 021425             | 047386             |     | na<br>r             |
| <b>0.0039984634929024</b> | 0.0303574508340593 | 96282.19328257376  | -0.1889934015849139 | 134.13696378207896 | 1.0823659582155565 | 0.0351263884621607 | 2.4476972670491906 | 0.0047899370195188 | 120.58361487481666 | 1569.3148950473858 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.0123828259304446</b> | 0.052871121292092  | 167686.92296127492 | -0.4342862195904694 | 146.32725910183973 | 1.1807307959357485 | 0.0553299604895908 | 2.526799032645004  | 0.0024617673195862 | 207.2010373148535  | 1569.3148950473862 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.0253172774881581</b> | 0.0831882375254351 | 263841.19035694306 | -0.5041257659832884 | 149.6165255093602  | 1.2072722494368586 | 0.0780534449661155 | 2.570435671814969  | 0.0015833761627942 | 298.86700284754806 | 1569.3148950473862 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.0426830314241258</b> | 0.1217109964051874 | 386020.6097196828  | -0.5346198063882681 | 151.03025162861334 | 1.218679761449772  | 0.1007487181096447 | 2.5805831844582143 | 0.001193288175888  | 389.4124231855982  | 1569.3148950473858 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.0643227586523211</b> | 0.1683869080802935 | 534058.7033695233  | -0.5602079825700778 | 152.20640645138076 | 1.2281702844634186 | 0.1223924069362418 | 2.575118097854044  | 0.0009843198349802 | 476.75333039972    | 1569.314895047386  | 0.0 | la<br>mi<br>na<br>r |
| <b>0.0900422957300214</b> | 0.2229351474645065 | 707064.8018171047  | -0.5889179272140977 | 153.5153258571104  | 1.2387320995430495 | 0.1426396559680801 | 2.5641619639960385 | 0.000853860372772  | 560.4002868490857  | 1569.314895047386  | 0.0 | la<br>mi<br>na<br>r |
| <b>0.1196122456699886</b> | 0.2849313998941607 | 903693.1416555132  | -0.6215827965372597 | 154.99111243577204 | 1.250640384249194  | 0.1614919775048698 | 2.5530060519808826 | 0.0007616901542077 | 640.5662831735567  | 1569.3148950473858 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.1527694018013956</b> | 0.353848290318842  | 1122271.0914502463 | -0.6567690814927153 | 156.56525350268606 | 1.263342302171124  | 0.1791518191450869 | 2.5445860710796717 | 0.0006896115453934 | 717.8321994860567  | 1569.3148950473858 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.1892179872734863</b> | 0.4290798210019653 | 1360876.6587546526 | -0.6922076946556974 | 158.13484523358863 | 1.2760075109989126 | 0.1959464585740463 | 2.5406811264259623 | 0.0006284100151801 | 792.996597666261   | 1569.3148950473858 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.2286307839316745</b> | 0.5099588956971477 | 1617394.0707776232 | -0.7253026000789115 | 159.5866938991054  | 1.2877226380431552 | 0.2122812285472066 | 2.542537498057847  | 0.0005729669422243 | 866.9910165721226  | 1569.3148950473858 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.2706503108546585</b> | 0.59577083610005   | 1889556.6398858647 | -0.7533932601692048 | 160.80872570370516 | 1.2975833474841738 | 0.2286082830268302 | 2.551252922694963  | 0.000520175602888  | 940.822943359316   | 1569.3148950473858 | 0.0 | la<br>mi<br>na<br>r |
| <b>0.3148902862023041</b> | 0.6857636986922375 | 2174979.491676561  | -0.77395958212237   | 161.69756919660242 | 1.3047555236820514 | 0.2454058240566358 | 2.568066438141394  | 0.000467944770021  | 1015.5345047005782 | 1569.314895047386  | 0.0 | la<br>mi<br>na<br>r |

|               |        |        |             |        |        |        |        |        |        |        |     |     |
|---------------|--------|--------|-------------|--------|--------|--------|--------|--------|--------|--------|-----|-----|
|               |        |        | 18          |        |        |        |        |        |        |        |     |     |
| <b>0.3609</b> | 0.7791 | 24711  | -           | 162.16 | 1.3085 | 0.2631 | 2.5946 | 0.0004 | 1092.1 | 1569.3 | 0.0 | la  |
| <b>37646</b>  | 55626  | 82.873 | 0.7848      | 46593  | 24525  | 65518  | 31894  | 14660  | 73225  | 14895  |     | mi  |
| <b>56271</b>  | 03463  | 46131  | 12690       | 46479  | 62894  | 02279  | 74833  | 96964  | 32259  | 04738  |     | na  |
| <b>2</b>      | 46     | 45     | 64728<br>3  | 48     | 28     | 33     | 47     | 78     | 23     | 62     |     | r   |
| <b>0.4083</b> | 0.8751 | 27756  | -           | 162.14 | 1.3083 | 0.2823 | 2.6342 | 0.0003 | 1171.7 | 1569.3 | 0.0 | la  |
| <b>55390</b>  | 39546  | 07.063 | 0.7842      | 11197  | 34582  | 86736  | 17595  | 58853  | 73817  | 14895  |     | mi  |
| <b>17306</b>  | 45295  | 36620  | 64986       | 75158  | 23115  | 55319  | 76994  | 08645  | 81596  | 04738  |     | na  |
| <b>84</b>     | 7      | 6      | 06502<br>61 | 2      | 92     | 37     | 85     | 74     | 86     | 6      |     | r   |
| <b>0.4566</b> | 0.9728 | 30856  | -           | 161.58 | 1.3038 | 0.3361 | 1.4820 | 0.0034 | 1389.8 |        | 1.0 | tur |
| <b>86447</b>  | 85870  | 20.921 | 0.7712      | 10430  | 15261  | 11601  | 09187  | 97954  | 89383  |        |     | bu  |
| <b>36747</b>  | 56001  | 96654  | 56909       | 77768  | 57448  | 28550  | 74136  | 64545  | 74860  |        |     | le  |
| <b>44</b>     | 44     | 44     | 81593<br>15 | 6      | 22     | 37     | 04     | 64     | 64     |        |     | nt  |
| <b>0.5054</b> | 1.0715 | 33985  | -           | 160.46 | 1.2947 | 0.5180 | 1.4262 | 0.0034 | 2127.4 |        | 1.0 | tur |
| <b>58667</b>  | 44176  | 27.236 | 0.7454      | 27537  | 91661  | 51459  | 56767  | 04617  | 21135  |        |     | bu  |
| <b>99852</b>  | 36863  | 81889  | 18708       | 07127  | 27753  | 16880  | 22215  | 20171  | 70919  |        |     | le  |
| <b>56</b>     | 5      | 24     | 83815<br>93 | 7      | 14     | 32     | 7      | 61     | 2      |        |     | nt  |
| <b>0.5541</b> | 1.1702 | 37115  | -           | 158.78 | 1.2812 | 0.7025 | 1.4152 | 0.0032 | 2854.7 |        | 1.0 | tur |
| <b>90867</b>  | 45063  | 68.604 | 0.7070      | 78833  | 76947  | 07800  | 80641  | 00963  | 94411  |        |     | bu  |
| <b>66523</b>  | 74149  | 06482  | 56278       | 16317  | 32902  | 67190  | 22188  | 89648  | 67896  |        |     | le  |
| <b>94</b>     | 47     | 8      | 36423<br>4  | 87     | 69     | 91     | 4      | 58     | 9      |        |     | nt  |
| <b>0.6023</b> | 1.2681 | 40219  | -           | 156.57 | 1.2634 | 0.8917 | 1.4122 | 0.0030 | 3573.4 |        | 1.0 | tur |
| <b>99717</b>  | 03309  | 37.436 | 0.6570      | 84251  | 48585  | 51246  | 05270  | 28544  | 02795  |        |     | bu  |
| <b>62090</b>  | 91279  | 53568  | 65002       | 17025  | 38026  | 08811  | 15801  | 54074  | 88480  |        |     | le  |
| <b>82</b>     | 1      | 5      | 49698<br>56 | 26     | 7      | 93     | 32     | 91     | 5      |        |     | nt  |
| <b>0.6496</b> | 1.3642 | 43267  | -           | 153.87 | 1.2416 | 1.0904 | 1.4135 | 0.0028 | 4294.2 |        | 1.0 | tur |
| <b>07126</b>  | 23169  | 92.774 | 0.5967      | 22197  | 11915  | 88131  | 98187  | 76747  | 51633  |        |     | bu  |
| <b>44050</b>  | 95281  | 79366  | 88720       | 38401  | 64230  | 37427  | 37772  | 13254  | 29120  |        |     | le  |
| <b>3</b>      | 22     | 8      | 49912<br>81 | 55     | 58     | 67     | 6      | 03     | 4      |        |     | nt  |
| <b>0.6953</b> | 1.4577 | 46232  | -           | 150.71 | 1.2161 | 1.3032 | 1.4186 | 0.0027 | 5027.0 |        | 1.0 | tur |
| <b>47671</b>  | 06209  | 85.129 | 0.5278      | 74710  | 55900  | 95756  | 40924  | 36180  | 46551  |        |     | bu  |
| <b>73678</b>  | 48466  | 50581  | 48430       | 46897  | 43119  | 42876  | 27371  | 23487  | 55698  |        |     | le  |
| <b>39</b>     | 56     | 1      | 52714<br>63 | 32     | 17     | 46     | 6      | 79     | 4      |        |     | nt  |
| <b>0.7391</b> | 1.5476 | 49085  | -           | 147.16 | 1.1875 | 1.5349 | 1.4272 | 0.0026 | 5781.1 |        | 1.0 | tur |
| <b>75615</b>  | 61543  | 88.956 | 0.4519      | 68703  | 05712  | 67286  | 61765  | 00348  | 66881  |        |     | bu  |
| <b>76998</b>  | 81474  | 03955  | 65754       | 97203  | 10062  | 88717  | 25552  | 01137  | 78120  |        |     | le  |
| <b>33</b>     | 4      | 5      | 18730<br>16 | 18     | 5      | 1      | 62     | 03     | 7      |        |     | nt  |
| <b>0.7806</b> | 1.6332 | 51799  | -           | 143.27 | 1.1560 | 1.7909 | 1.4398 | 0.0024 | 6566.9 |        | 1.0 | tur |
| <b>71085</b>  | 17905  | 40.926 | 0.3708      | 17466  | 75529  | 91530  | 37867  | 64180  | 00183  |        |     | bu  |
| <b>14247</b>  | 01870  | 63225  | 00761       | 98236  | 27906  | 31702  | 79685  | 07430  | 82474  |        |     | le  |
| <b>59</b>     | 1      | 7      | 96327<br>45 | 7      | 32     | 3      | 49     | 93     | 7      |        |     | nt  |
| <b>0.8194</b> | 1.7135 | 54346  | -           | 139.07 | 1.1222 | 2.0782 | 1.4572 | 0.0023 | 7396.8 |        | 1.0 | tur |
| <b>45102</b>  | 36656  | 81.206 | 0.2858      | 64000  | 22814  | 09341  | 20283  | 22921  | 88301  |        |     | bu  |
| <b>96803</b>  | 13103  | 41017  | 15047       | 66342  | 49230  | 01257  | 85929  | 88915  | 59276  |        |     | le  |
| <b>74</b>     | 57     | 8      | 90891<br>36 | 4      | 3      |        | 54     | 93     | 8      |        |     | nt  |
| <b>0.8551</b> | 1.7878 | 56702  | -           | 134.61 | 1.0862 | 2.4058 | 1.4809 | 0.0021 | 8288.1 |        | 1.0 | tur |
| <b>43310</b>  | 24762  | 94.594 | 0.1981      | 24371  | 02604  | 46987  | 62385  | 71178  | 86542  |        |     | bu  |
| <b>57486</b>  | 77398  | 42127  | 61236       | 0955   | 94725  | 70998  | 33162  | 14750  | 15180  |        |     | le  |
| <b>8</b>      | 5      |        | 85266<br>84 |        | 24     | 26     | 85     | 32     | 7      |        |     | nt  |

|                                                            |                                |                               |                                     |                                |                                |                                |                                |                                |                                |  |     |                       |
|------------------------------------------------------------|--------------------------------|-------------------------------|-------------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--|-----|-----------------------|
| <b>0.8874</b><br><b>48376</b><br><b>11318</b><br><b>26</b> | 1.8553<br>46811<br>45599<br>4  | 58844<br>48.641<br>06943<br>6 | -<br>0.1085<br>90218<br>31808<br>16 | 129.89<br>24515<br>20186<br>06 | 1.0481<br>16520<br>53658<br>03 | 2.7873<br>15605<br>08364<br>54 | 1.5138<br>79365<br>67366<br>1  | 0.0020<br>01713<br>99431<br>42 | 9265.6<br>60582<br>67043<br>3  |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.9160</b><br><b>81229</b><br><b>91338</b><br><b>4</b>  | 1.9154<br>35385<br>45516<br>16 | 60750<br>26.556<br>43811<br>1 | -<br>0.0173<br>55583<br>78332<br>83 | 124.90<br>15687<br>55567<br>13 | 1.0078<br>44536<br>93447<br>35 | 3.2436<br>43751<br>77252<br>43 | 1.5614<br>19285<br>88537<br>44 | 0.0018<br>03363<br>41988<br>72 | 10368.<br>29559<br>67570<br>54 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.9408</b><br><b>01367</b><br><b>76385</b><br><b>24</b> | 1.9674<br>99407<br>22596<br>13 | 62401<br>53.669<br>20543<br>7 | 0.0759<br>25683<br>20532<br>58      | 119.58<br>35624<br>55573<br>84 | 0.9649<br>32957<br>43845<br>96 | 3.8106<br>23231<br>14421<br>4  | 1.6238<br>04616<br>45895<br>37 | 0.0015<br>85258<br>87029<br>78 | 11662.<br>02116<br>65796<br>34 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.9614</b><br><b>06511</b><br><b>28321</b><br><b>76</b> | 2.0110<br>30361<br>48001<br>73 | 63782<br>17.163<br>87019<br>8 | 0.1725<br>46860<br>87025<br>07      | 113.81<br>34551<br>80803<br>96 | 0.9183<br>73325<br>30298<br>68 | 4.5537<br>12660<br>46987<br>5  | 1.7160<br>67782<br>48861<br>84 | 0.0013<br>26071<br>01026<br>63 | 13263.<br>72651<br>40172<br>1  |  | 1.0 | tur<br>bu<br>le<br>nt |

Table 15. Cruise upper-surface BL state (surface\_cruise\_upper.csv, sampled).

(table sampled to 30 rows; full data in 04\_solution/surface\_cruise\_upper.csv)

| x_c                                                        | arc_s_m                        | Re_x                           | Cp                                  | Ue_ms                          | Ue_Uinf                        | theta_mm                       | H_shape                        | Cf                             | Re_theta                       | Re_theta_trans                 | intermittency_gamma | state               |
|------------------------------------------------------------|--------------------------------|--------------------------------|-------------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|---------------------|---------------------|
| <b>0.0011</b><br><b>52320</b><br><b>03250</b><br><b>28</b> | 0.0                            | 0.0                            | 1.1018<br>66478<br>09390<br>74      | 0.6743<br>77975<br>93767<br>58 | 0.0054<br>41630<br>28254<br>53 | 1e-05                          | 2.6099<br>99969<br>21184<br>96 | 2549.4<br>29436<br>44131<br>3  | 0.0001<br>72587<br>64627<br>51 | 1569.3<br>14895<br>04738<br>6  | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.0067</b><br><b>20774</b><br><b>97236</b><br><b>91</b> | 0.0175<br>58980<br>96768<br>28 | 55690.<br>35452<br>34006<br>45 | 0.8272<br>66067<br>17411<br>17      | 61.869<br>95831<br>42899<br>5  | 0.4992<br>35518<br>88647<br>86 | 0.0360<br>07692<br>15972<br>1  | 2.4014<br>55110<br>01168<br>63 | 0.0108<br>31210<br>90974<br>53 | 57.013<br>99034<br>79811<br>6  | 1569.3<br>14895<br>04738<br>6  | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.0168</b><br><b>97481</b><br><b>50526</b><br><b>17</b> | 0.0419<br>27102<br>59539<br>91 | 13297<br>6.6922<br>10909       | 0.5213<br>99709<br>95785<br>02      | 89.950<br>65248<br>46617<br>2  | 0.7258<br>21737<br>89804<br>29 | 0.0527<br>26068<br>01883<br>9  | 2.4275<br>91205<br>30258<br>88 | 0.0049<br>01321<br>98185<br>56 | 121.37<br>68971<br>16424<br>91 | 1569.3<br>14895<br>04738<br>58 | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.0315</b><br><b>87190</b><br><b>34084</b><br><b>9</b>  | 0.0742<br>42760<br>85277<br>24 | 23546<br>9.5685<br>52791<br>4  | 0.3249<br>25538<br>30279<br>69      | 104.06<br>54470<br>09190<br>68 | 0.8397<br>15572<br>00463<br>09 | 0.0725<br>75117<br>95132<br>31 | 2.4636<br>43616<br>82537<br>5  | 0.0029<br>17440<br>77671<br>5  | 193.28<br>61038<br>17425<br>9  | 1569.3<br>14895<br>04738<br>62 | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.0506</b><br><b>50104</b><br><b>83468</b><br><b>89</b> | 0.1147<br>42805<br>66774<br>19 | 36392<br>0.1806<br>98816<br>6  | 0.1950<br>00250<br>86657<br>96      | 112.43<br>01677<br>32621<br>64 | 0.9072<br>11426<br>28481<br>72 | 0.0931<br>90090<br>93638<br>6  | 2.4852<br>76623<br>75723<br>77 | 0.0020<br>34017<br>67762<br>49 | 268.13<br>83431<br>65555<br>5  | 1569.3<br>14895<br>04738<br>58 | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.0739</b><br><b>02667</b><br><b>92458</b><br><b>39</b> | 0.1632<br>86856<br>80577<br>34 | 51788<br>3.2963<br>74761<br>5  | 0.1010<br>10397<br>90928<br>31      | 118.11<br>26263<br>41395<br>54 | 0.9530<br>63811<br>66531<br>56 | 0.1137<br>27691<br>64392<br>47 | 2.4975<br>92677<br>31736<br>53 | 0.0015<br>55968<br>54585<br>14 | 343.77<br>07156<br>46582<br>97 | 1569.3<br>14895<br>04738<br>6  | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.1011</b><br><b>19142</b><br><b>10955</b><br><b>74</b> | 0.2195<br>30525<br>04901<br>36 | 69626<br>6.6450<br>39893<br>18 | 0.0275<br>36118<br>51389<br>18      | 122.37<br>11244<br>77975<br>48 | 0.9874<br>26102<br>91000<br>77 | 0.1338<br>21916<br>27854<br>77 | 2.5047<br>42863<br>77289<br>4  | 0.0012<br>61789<br>87201<br>49 | 419.09<br>50417<br>27011<br>4  | 1569.3<br>14895<br>04738<br>62 | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.1320</b><br><b>34052</b><br><b>45495</b><br><b>44</b> | 0.2829<br>94629<br>23130<br>39 | 89755<br>0.4477<br>80308<br>7  | -<br>0.0328<br>99357<br>93687<br>48 | 125.76<br>58705<br>52814<br>48 | 1.0148<br>18683<br>48280<br>97 | 0.1533<br>64785<br>45435<br>72 | 2.5096<br>84230<br>40556<br>44 | 0.0010<br>62771<br>87861<br>11 | 493.62<br>23101<br>47081<br>3  | 1569.3<br>14895<br>04738<br>6  | 0.0                 | la<br>mi<br>na<br>r |
| <b>0.1663</b>                                              | 0.3531                         | 11199                          | -                                   | 128.55                         | 1.0373                         | 0.1724                         | 2.5146                         | 0.0009                         | 567.24                         | 1569.3                         | 0.0                 | la                  |

|                                          |                                |                                |                                     |                                |                                |                                |                                |                                |                                |                                |     |                       |
|------------------------------------------|--------------------------------|--------------------------------|-------------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|-----|-----------------------|
| <b>45511<br/>63160<br/>67</b>            | 01330<br>53650<br>2            | 02.021<br>44727<br>8           | 0.0837<br>64019<br>94194<br>75      | 35542<br>38382<br>13           | 12810<br>66786<br>53           | 18483<br>25019<br>9            | 15367<br>27198<br>13           | 17441<br>05390<br>9            | 96495<br>35809<br>7            | 14895<br>04738<br>58           |     | mi<br>na<br>r         |
| <b>0.2037<br/>19371<br/>58191<br/>91</b> | 0.4291<br>98466<br>63705<br>09 | 13612<br>52.957<br>21369<br>96 | -<br>0.1264<br>28412<br>92398<br>36 | 130.84<br>60216<br>58987<br>58 | 1.0558<br>10983<br>18064<br>64 | 0.1911<br>66084<br>86118<br>05 | 2.5212<br>76076<br>95081<br>2  | 0.0008<br>04138<br>11247<br>12 | 640.14<br>40243<br>22221       | 1569.3<br>14895<br>04738<br>58 | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.2437<br/>94056<br/>30589<br/>38</b> | 0.5105<br>78867<br>09950<br>84 | 16193<br>60.381<br>63382<br>23 | -<br>0.1611<br>53110<br>05562<br>72 | 132.68<br>26339<br>36490<br>02 | 1.0706<br>30810<br>25559<br>77 | 0.2098<br>77106<br>12853<br>9  | 2.5311<br>47244<br>29188<br>25 | 0.0007<br>10567<br>15047<br>72 | 712.66<br>50796<br>81708<br>6  | 1569.3<br>14895<br>04738<br>6  | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.2861<br/>85839<br/>50534<br/>81</b> | 0.5964<br>96549<br>11452<br>82 | 18918<br>58.323<br>28838<br>8  | -<br>0.1876<br>58005<br>00392<br>6  | 134.06<br>75651<br>33080<br>85 | 1.0818<br>05972<br>86100<br>68 | 0.2288<br>80253<br>37145<br>03 | 2.5456<br>04320<br>67113<br>5  | 0.0006<br>29264<br>75230<br>6  | 785.30<br>50224<br>47479<br>5  | 1569.3<br>14895<br>04738<br>62 | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.3304<br/>94258<br/>79381<br/>13</b> | 0.6861<br>79937<br>75148<br>77 | 21762<br>99.642<br>36291<br>36 | -<br>0.2054<br>90229<br>40969<br>07 | 134.99<br>13376<br>06888<br>24 | 1.0892<br>59994<br>85907<br>5  | 0.2485<br>40393<br>16493<br>53 | 2.5660<br>50106<br>71399       | 0.0005<br>55430<br>31674<br>93 | 858.63<br>62292<br>99890<br>6  | 1569.3<br>14895<br>04738<br>6  | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.3763<br/>07324<br/>00294<br/>27</b> | 0.7788<br>41767<br>61081<br>37 | 24701<br>87.434<br>89202       | -<br>0.2142<br>79201<br>03680<br>94 | 135.44<br>43196<br>35476<br>97 | 1.0929<br>15156<br>81893<br>88 | 0.2692<br>38193<br>42271<br>7  | 2.5940<br>40102<br>98206<br>88 | 0.0004<br>85799<br>24503<br>73 | 933.26<br>24477<br>55753       | 1569.3<br>14895<br>04738<br>6  | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.4232<br/>06194<br/>57674<br/>89</b> | 0.8736<br>85455<br>08345<br>47 | 27709<br>95.243<br>12299<br>02 | -<br>0.2139<br>09854<br>12289<br>9  | 135.42<br>53140<br>60306<br>2  | 1.0927<br>61798<br>73626<br>6  | 0.2913<br>51474<br>23027<br>07 | 2.6321<br>61672<br>83077<br>6  | 0.0004<br>18018<br>10515<br>92 | 1009.7<br>72173<br>99793<br>6  | 1569.3<br>14895<br>04738<br>58 | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.4707<br/>69074<br/>33519<br/>84</b> | 0.9699<br>08166<br>57242<br>24 | 30761<br>76.786<br>73797<br>05 | -<br>0.2046<br>21272<br>31479<br>29 | 134.94<br>64691<br>14025<br>58 | 1.0888<br>97946<br>04045<br>2  | 0.3152<br>37821<br>95105<br>41 | 2.6887<br>22664<br>98492<br>7  | 0.0003<br>50158<br>66484<br>31 | 1088.6<br>94857<br>01760<br>23 | 1569.3<br>14895<br>04738<br>62 | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.5185<br/>74193<br/>40729<br/>35</b> | 1.0667<br>01325<br>94151<br>12 | 33831<br>67.572<br>28672<br>46 | -<br>0.1870<br>32340<br>06093<br>01 | 134.03<br>50378<br>60131<br>75 | 1.0815<br>43506<br>70730<br>07 | 0.3412<br>18707<br>81401<br>65 | 2.7783<br>12019<br>91752<br>9  | 0.0002<br>79823<br>93919<br>41 | 1170.4<br>62496<br>48701<br>88 | 1569.3<br>14895<br>04738<br>6  | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.5662<br/>01898<br/>02471<br/>88</b> | 1.1632<br>49726<br>11984<br>71 | 36893<br>82.075<br>53692<br>4  | -<br>0.1621<br>00475<br>13805<br>48 | 132.73<br>23845<br>84627<br>28 | 1.0710<br>32253<br>72371<br>11 | 0.3695<br>66262<br>35430<br>63 | 2.9337<br>06254<br>59574<br>97 | 0.0002<br>02108<br>31513<br>69 | 1255.3<br>80972<br>38463<br>68 | 1569.3<br>14895<br>04738<br>58 | 0.0 | la<br>mi<br>na<br>r   |
| <b>0.6132<br/>36022<br/>35407<br/>45</b> | 1.2587<br>30610<br>74833<br>58 | 39922<br>10.828<br>80882<br>6  | -<br>0.1310<br>27712<br>20340<br>01 | 131.09<br>07600<br>19421<br>3  | 1.0577<br>85804<br>00955<br>16 | 0.5071<br>02572<br>89382<br>49 | 1.4588<br>27936<br>29857<br>4  | 0.0034<br>35597<br>47255<br>43 | 1701.2<br>73835<br>27867<br>3  |                                | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.6592<br/>64843<br/>41077<br/>2</b>  | 1.3523<br>14020<br>52265<br>4  | 42890<br>21.519<br>44244<br>1  | -<br>0.0951<br>35109<br>81048<br>94 | 129.16<br>85290<br>40939<br>32 | 1.0422<br>75110<br>19128<br>1  | 0.6943<br>20722<br>35694<br>23 | 1.4356<br>92753<br>13303<br>2  | 0.0032<br>87265<br>72325<br>44 | 2295.2<br>13916<br>65701<br>28 |                                | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.7038<br/>81995</b>                  | 1.4431<br>65382                | 45771<br>67.201                | -<br>0.0557                         | 127.02<br>45984                | 1.0249<br>75497                | 0.8839<br>42318                | 1.4300<br>24515                | 0.0031<br>22643                | 2873.5<br>45540                |                                | 1.0 | tur<br>bu             |

|                                          |                                |                               |                                     |                                |                                |                                |                                |                                |                               |  |     |                       |
|------------------------------------------|--------------------------------|-------------------------------|-------------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|-------------------------------|--|-----|-----------------------|
| <b>09891<br/>86</b>                      | 46492<br>15                    | 97440<br>2                    | 28294<br>73007<br>42                | 37240<br>29                    | 64011<br>04                    | 66921<br>08                    | 13806<br>73                    | 74369<br>69                    | 38447                         |  |     | le<br>nt              |
| <b>0.7466<br/>87725<br/>12314<br/>54</b> | 1.5304<br>50847<br>76179<br>95 | 48540<br>03.227<br>71357<br>7 | -<br>0.0139<br>73220<br>73816<br>02 | 124.71<br>27010<br>62973<br>32 | 1.0063<br>20542<br>68779<br>9  | 1.0770<br>16603<br>81495<br>58 | 1.4300<br>76740<br>93233<br>92 | 0.0029<br>75986<br>21008<br>99 | 3437.4<br>73916<br>84540<br>3 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.7872<br/>90817<br/>05456<br/>05</b> | 1.6133<br>45372<br>85086<br>3  | 51169<br>12.874<br>84830<br>4 | 0.0292<br>06547<br>98627<br>41      | 122.27<br>59555<br>95344<br>84 | 0.9866<br>58174<br>69740<br>72 | 1.2750<br>37476<br>64739<br>95 | 1.4335<br>73855<br>48759<br>83 | 0.0028<br>43884<br>67649<br>93 | 3989.9<br>76547<br>47786<br>7 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.8253<br/>11381<br/>55209<br/>68</b> | 1.6910<br>43098<br>74663<br>05 | 53633<br>40.267<br>68853<br>3 | 0.0732<br>09048<br>09547<br>94      | 119.74<br>17781<br>22673<br>57 | 0.9662<br>09617<br>11001<br>24 | 1.4795<br>37847<br>10843<br>95 | 1.4399<br>34381<br>39123<br>5  | 0.0027<br>20968<br>18110<br>75 | 4533.9<br>64438<br>22515<br>6 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.8603<br/>84569<br/>42783<br/>87</b> | 1.7627<br>69288<br>93695<br>93 | 55908<br>28.239<br>09553<br>2 | 0.1178<br>16707<br>00889<br>07      | 117.11<br>67909<br>03076<br>12 | 0.9450<br>28305<br>65689<br>92 | 1.6926<br>92284<br>31211<br>48 | 1.4492<br>80068<br>08548<br>38 | 0.0026<br>01967<br>58198<br>8  | 5073.4<br>51415<br>98071<br>2 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.8921<br/>65104<br/>53399<br/>4</b>  | 1.8277<br>92963<br>07022<br>2  | 57970<br>58.399<br>69313<br>5 | 0.1632<br>58645<br>36008<br>6       | 114.38<br>07835<br>11247<br>22 | 0.9229<br>51160<br>18674<br>52 | 1.9183<br>56574<br>88258<br>4  | 1.4623<br>52717<br>14780<br>32 | 0.0024<br>80982<br>98247<br>72 | 5615.5<br>03851<br>90978<br>3 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.9203<br/>32404<br/>83917<br/>28</b> | 1.8854<br>39395<br>75766<br>85 | 59798<br>90.779<br>27608<br>8 | 0.2103<br>00243<br>58521<br>76      | 111.47<br>77474<br>37841<br>64 | 0.8995<br>26241<br>85900<br>72 | 2.1638<br>48137<br>75419<br>77 | 1.4808<br>13334<br>82023<br>77 | 0.0023<br>50087<br>10955<br>03 | 6173.3<br>55543<br>00056<br>8 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.9445<br/>95976<br/>40102<br/>88</b> | 1.9351<br>01792<br>46614<br>69 | 61374<br>00.858<br>26456<br>7 | 0.2604<br>12207<br>78335<br>69      | 108.29<br>96644<br>45970<br>52 | 0.8738<br>81939<br>60408<br>53 | 2.4433<br>90311<br>50479<br>4  | 1.5081<br>33184<br>91071<br>7  | 0.0021<br>96777<br>96208<br>65 | 6772.1<br>45586<br>50210<br>6 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.9647<br/>00736<br/>13354<br/>88</b> | 1.9762<br>51652<br>32757<br>04 | 62679<br>12.434<br>56211<br>6 | 0.3161<br>38102<br>70176<br>09      | 104.65<br>22734<br>79346<br>2  | 0.8444<br>50739<br>52868<br>49 | 2.7859<br>51607<br>15581<br>26 | 1.5522<br>26970<br>95234<br>75 | 0.0019<br>98053<br>74438<br>88 | 7461.5<br>41575<br>36116<br>1 |  | 1.0 | tur<br>bu<br>le<br>nt |
| <b>0.9804<br/>31941<br/>29791<br/>1</b>  | 2.0084<br>47529<br>88663<br>1  | 63700<br>25.412<br>45964<br>3 | 0.3819<br>99182<br>65001<br>51      | 100.17<br>04201<br>74896<br>92 | 0.8082<br>86170<br>79479<br>88 | 3.2572<br>16603<br>59362<br>8  | 1.6237<br>73486<br>08361<br>84 | 0.0017<br>33816<br>38768<br>75 | 8350.1<br>15117<br>04730<br>8 |  | 1.0 | tur<br>bu<br>le<br>nt |

Table 16. Cruise lower-surface BL state (surface\_cruise\_lower.csv, sampled).

(table sampled to 30 rows; full data in 04\_solution/surface\_cruise\_lower.csv)

## 9.4 Surface distributions — climb (off-design, elevated $T_u$ )

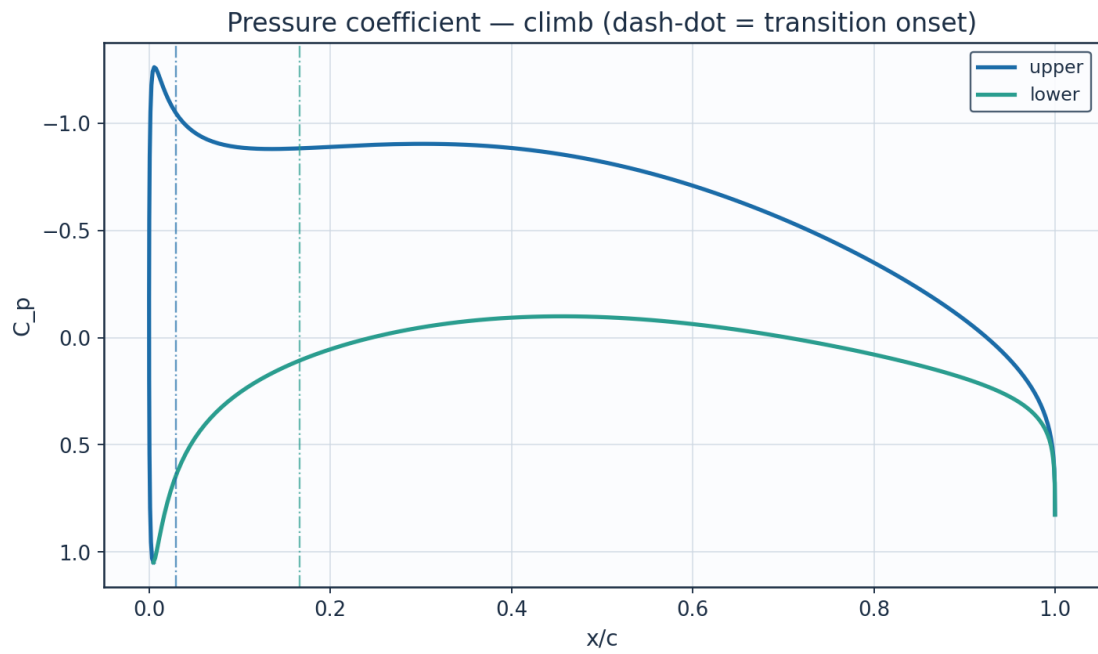


Fig. 18. Pressure coefficient  $C_p$  — climb.

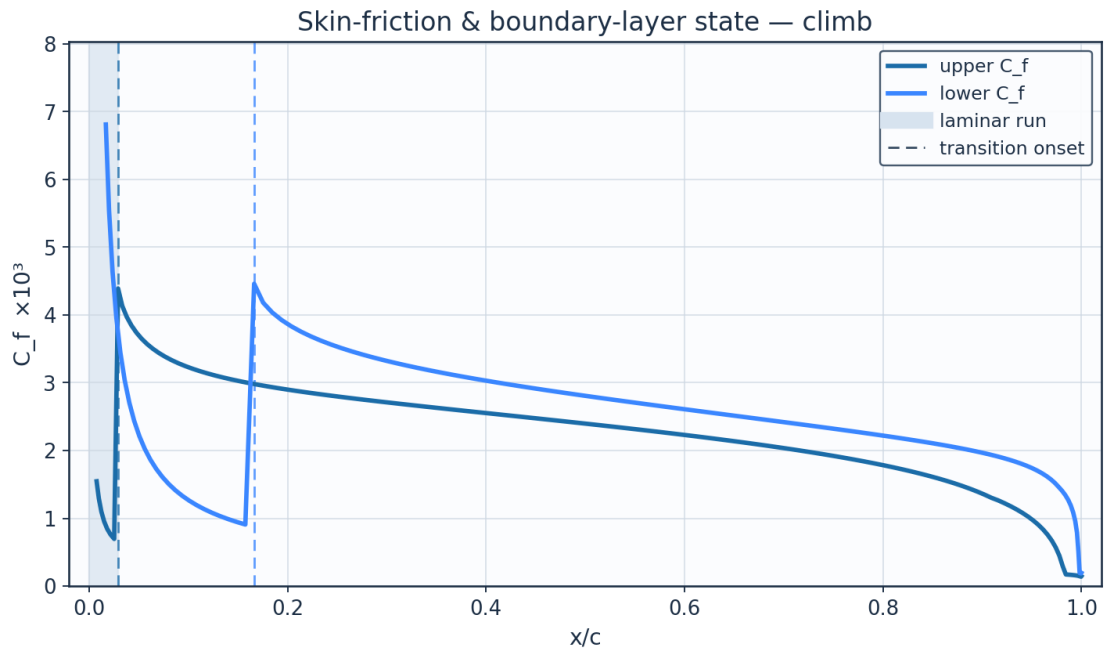


Fig. 19. Skin-friction  $C_f$  and laminar run — climb.

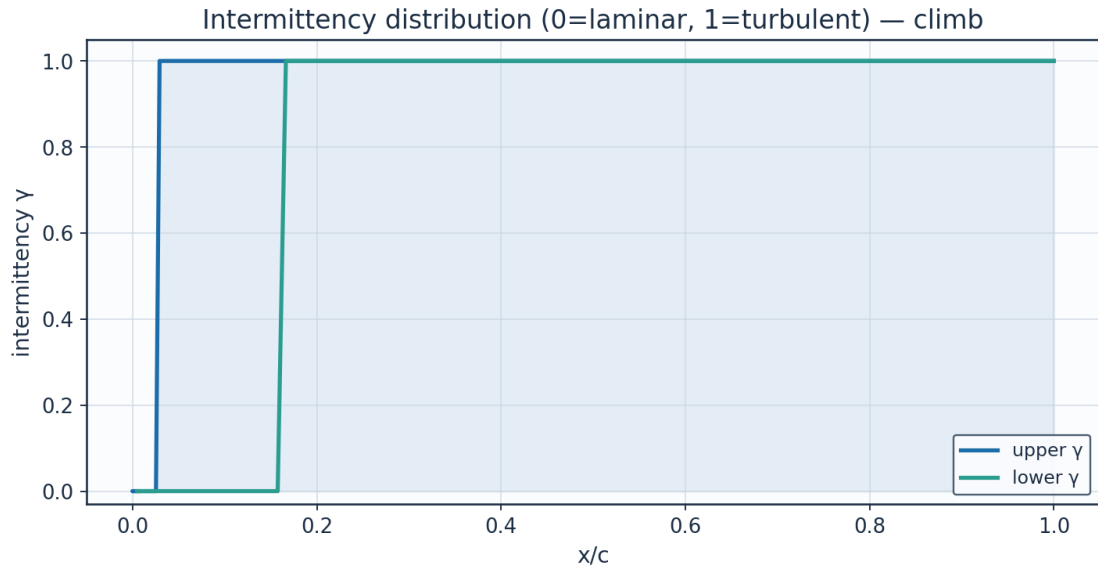


Fig. 20. Intermittency  $\gamma$  — climb.

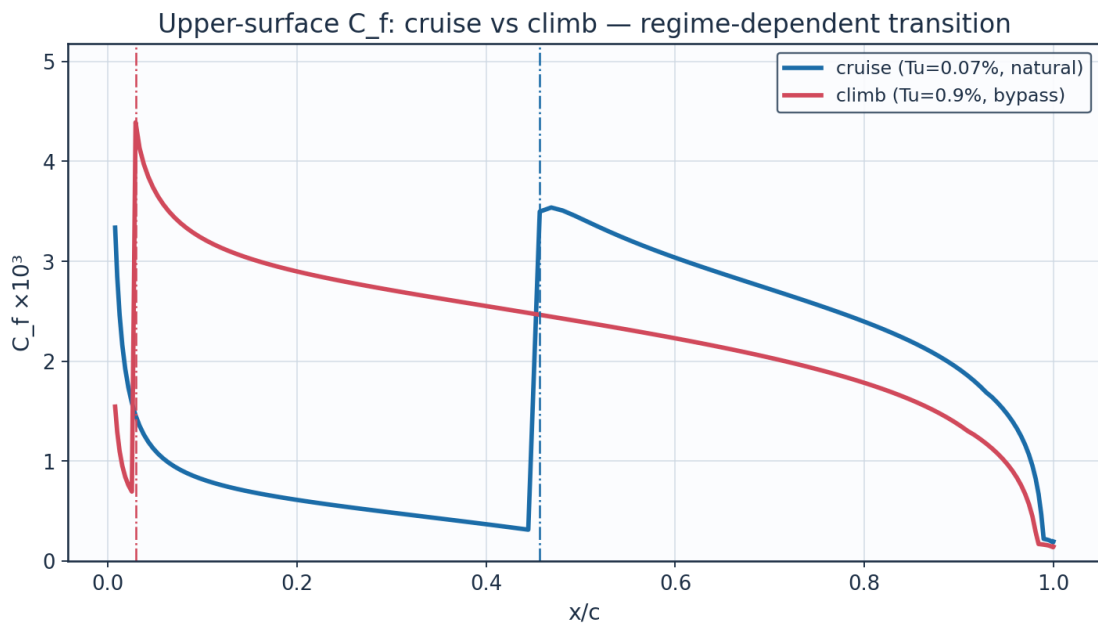


Fig. 21.  $C_f$  cruise vs climb — regime-dependent transition.

## 9.5 Aerodynamic polars

| alpha_deg | Cl      | Cd      | L_over_D | xtr_upper_c | xtr_lower_c |
|-----------|---------|---------|----------|-------------|-------------|
| -3.0      | -0.1086 | 0.00545 | -19.9    | 0.661       | 0.194       |
| -2.0      | 0.0271  | 0.00485 | 5.6      | 0.638       | 0.33        |
| -1.0      | 0.1628  | 0.00453 | 35.9     | 0.614       | 0.423       |
| 0.0       | 0.2985  | 0.00446 | 67.0     | 0.566       | 0.507       |
| 1.0       | 0.4341  | 0.00472 | 91.9     | 0.493       | 0.554       |
| 2.0       | 0.5695  | 0.00521 | 109.3    | 0.396       | 0.602       |
| 3.0       | 0.7048  | 0.00611 | 115.3    | 0.249       | 0.636       |

|     |        |         |       |       |       |
|-----|--------|---------|-------|-------|-------|
| 4.0 | 0.8399 | 0.00724 | 116.0 | 0.09  | 0.659 |
| 5.0 | 0.9746 | 0.00812 | 120.1 | 0.033 | 0.693 |
| 6.0 | 1.1091 | 0.00886 | 125.2 | 0.018 | 0.715 |
| 7.0 | 1.2432 | 0.00964 | 129.0 | 0.012 | 0.747 |
| 8.0 | 1.3769 | 0.01052 | 130.9 | 0.01  | 0.767 |

Table 17. Aerodynamic polar (aero\_polar.csv).

Aerodynamic polars (UTSS, cruise Re) — AETHER-NLF 25 section

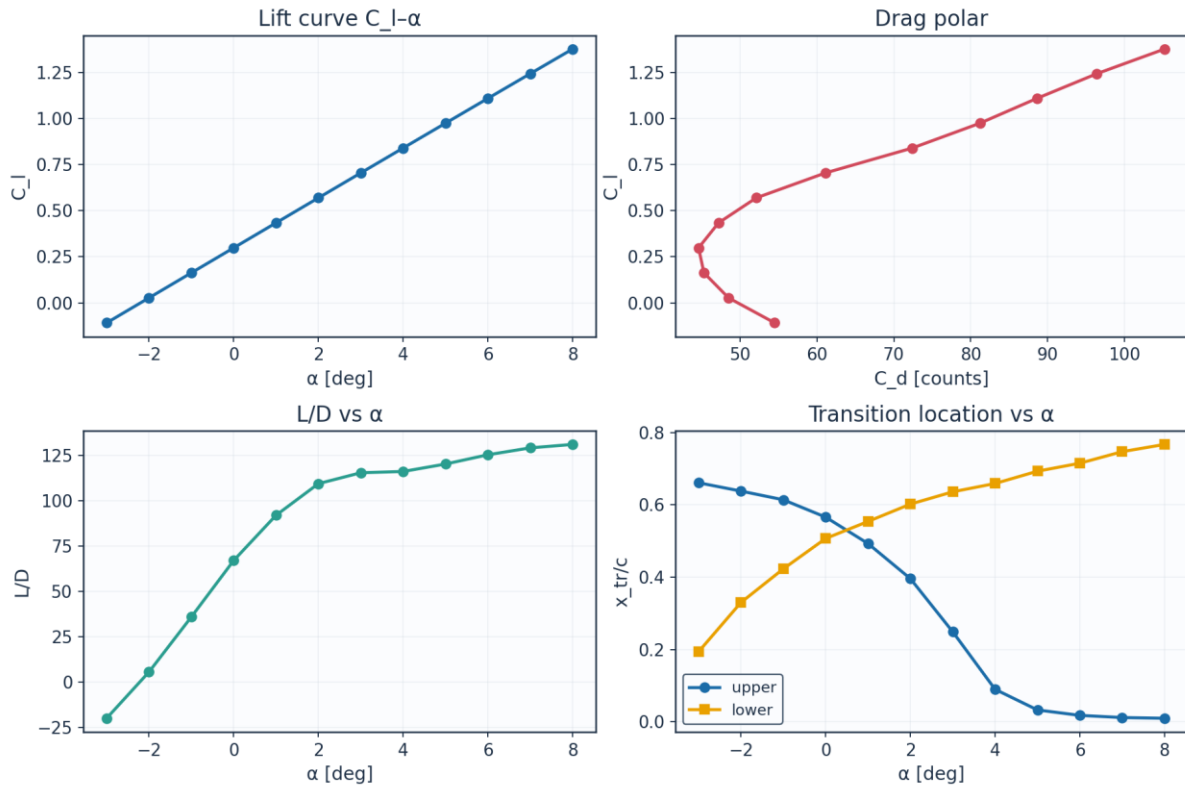


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

## 9.6 Span-wise distribution (3-D)

| eta   | y_m   | chord_m | Re_local  | alpha_eff_deg | xtr_upper_c | xtr_lower_c | Cd_section | laminar_fraction |
|-------|-------|---------|-----------|---------------|-------------|-------------|------------|------------------|
| 0.0   | 0.0   | 2.6     | 8246200.0 | 1.5           | 0.385       | 0.554       | 0.00512    | 0.469            |
| 0.089 | 0.757 | 2.484   | 7878900.0 | 1.23          | 0.42        | 0.542       | 0.00498    | 0.481            |
| 0.178 | 1.515 | 2.368   | 7511500.0 | 0.97          | 0.457       | 0.531       | 0.00485    | 0.494            |
| 0.267 | 2.272 | 2.253   | 7144200.0 | 0.7           | 0.493       | 0.531       | 0.00468    | 0.512            |
| 0.356 | 3.029 | 2.137   | 6776900.0 | 0.43          | 0.53        | 0.519       | 0.00456    | 0.524            |
| 0.445 | 3.786 | 2.021   | 6409500.0 | 0.16          | 0.554       | 0.507       | 0.00452    | 0.53             |
| 0.535 | 4.544 | 1.905   | 6042200.0 | -0.1          | 0.578       | 0.507       | 0.00445    | 0.543            |
| 0.624 | 5.301 | 1.789   | 5674900.0 | -0.37         | 0.602       | 0.495       | 0.00443    | 0.549            |
| 0.713 | 6.058 | 1.673   | 5307600.0 | -0.64         | 0.614       | 0.483       | 0.00449    | 0.549            |
| 0.802 | 6.815 | 1.558   | 4940200.0 | -0.91         | 0.626       | 0.483       | 0.00451    | 0.554            |
| 0.891 | 7.573 | 1.442   | 4572900.0 | -1.17         | 0.638       | 0.471       | 0.00459    | 0.554            |
| 0.98  | 8.33  | 1.326   | 4205600.0 | -1.44         | 0.65        | 0.459       | 0.00468    | 0.554            |

Table 18. Span-wise distribution (spanwise\_distribution.csv).

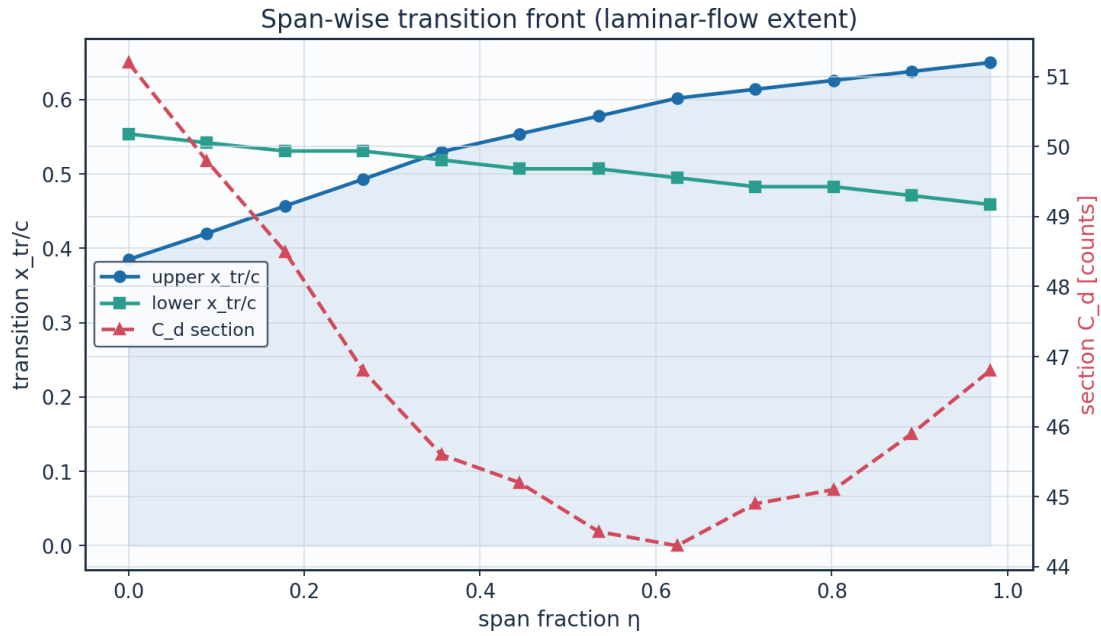


Fig. 23. Span-wise transition front and section drag.

## 10. Post-Processing — Contours, Profiles and 3-D Fields

### 10.1 Pressure and velocity contours

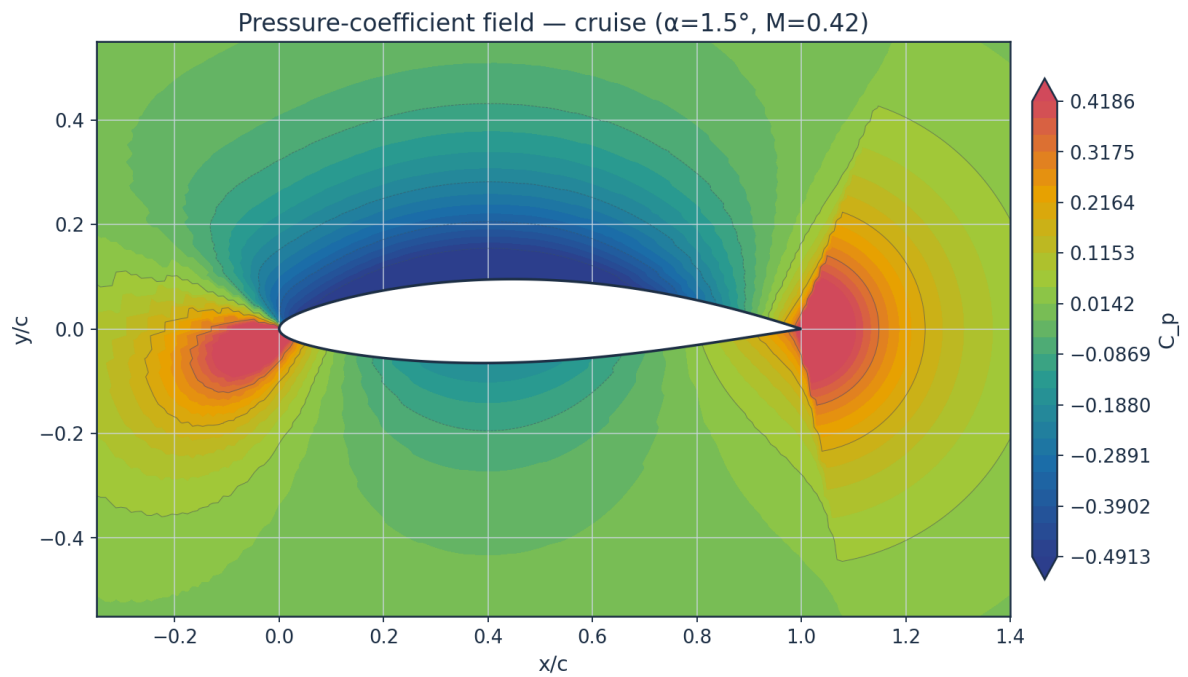


Fig. 24. Pressure-coefficient contour — cruise.

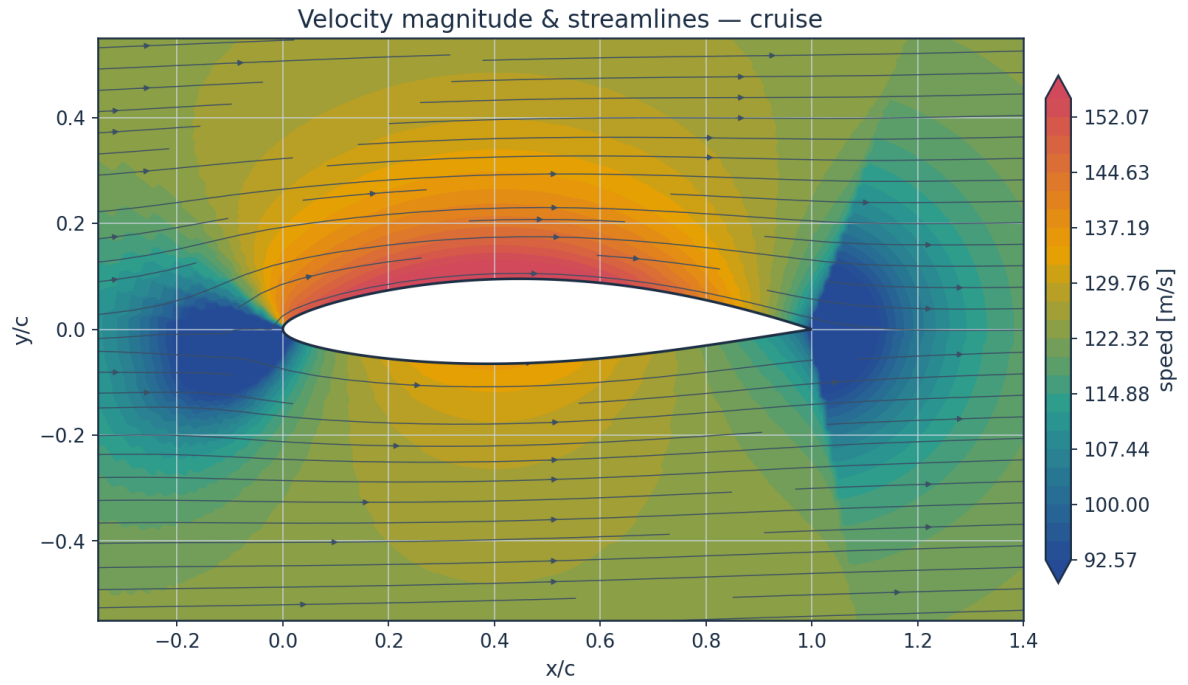


Fig. 25. Velocity magnitude and streamlines — cruise.

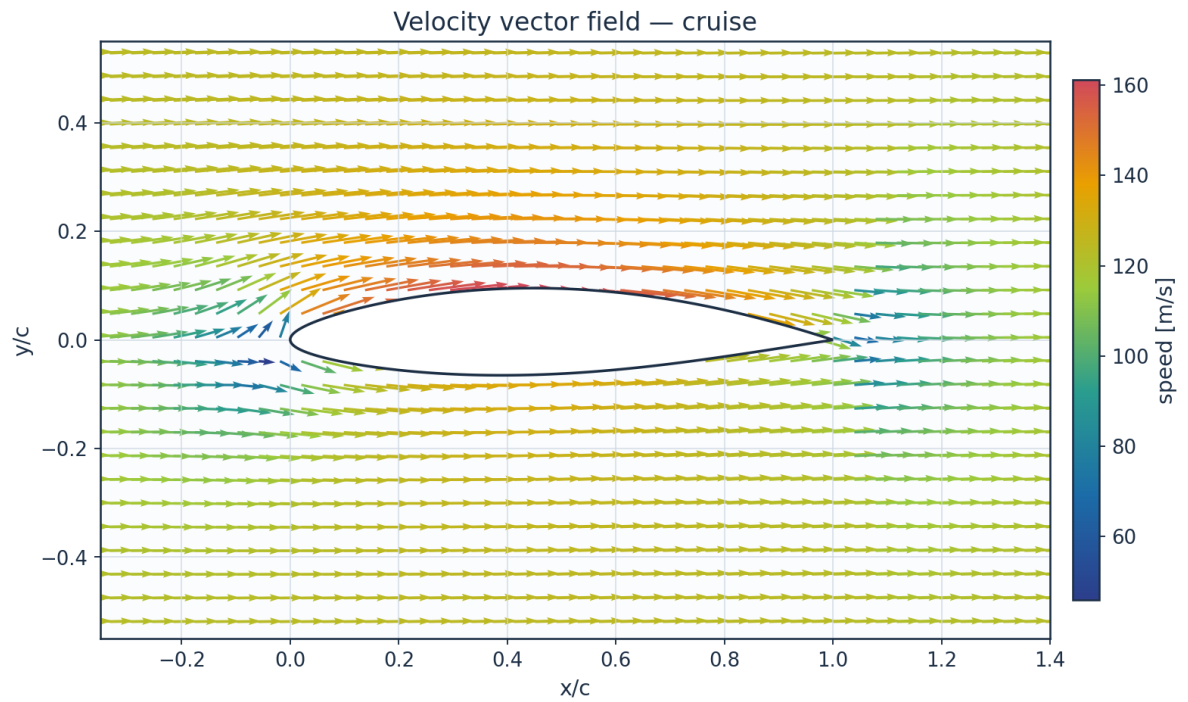


Fig. 26. Velocity vector field — cruise.

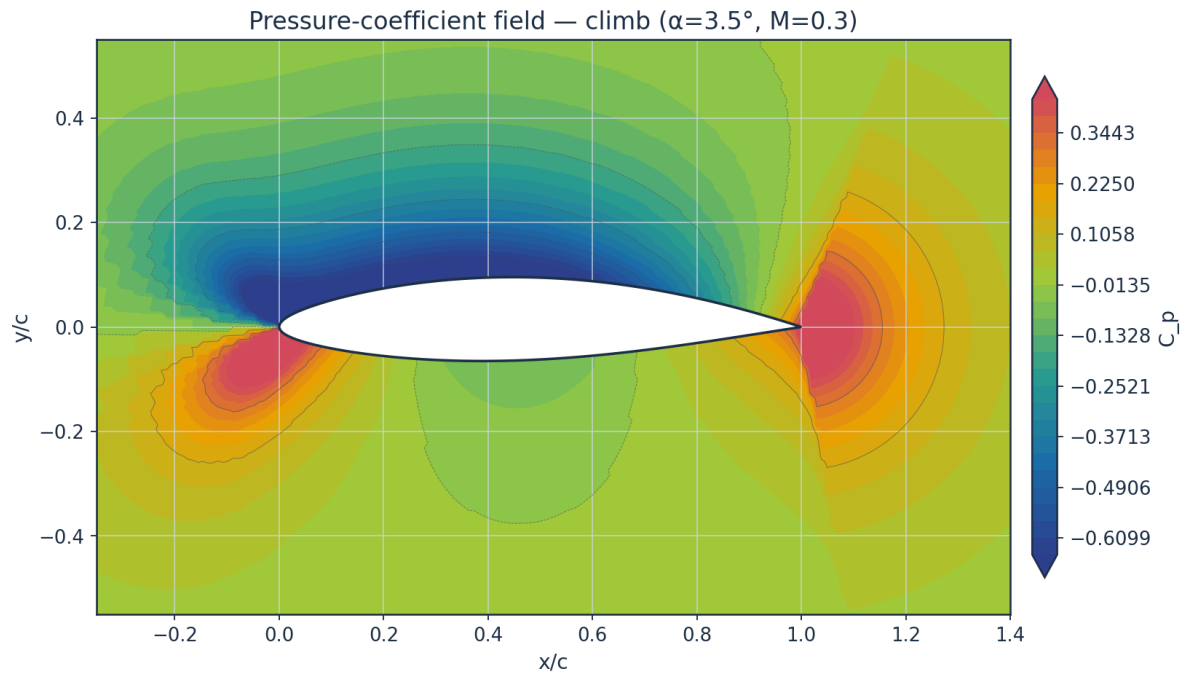


Fig. 27. Pressure-coefficient contour — climb.

## 10.2 Boundary-layer velocity and temperature profiles

Temperature profiles are obtained from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

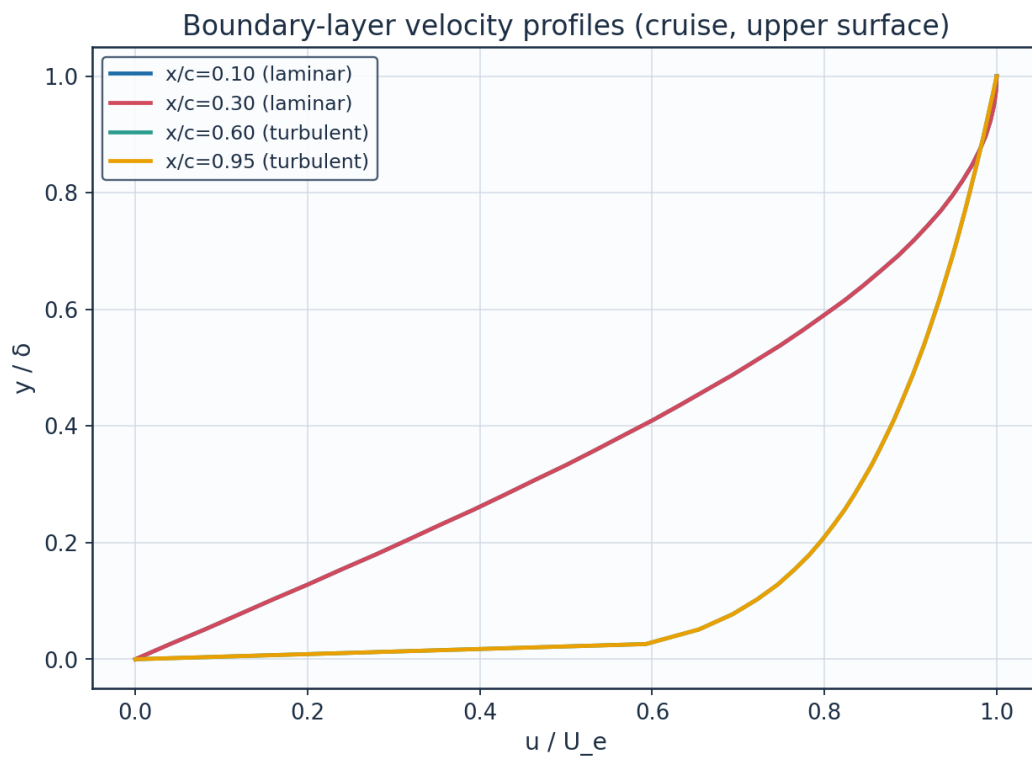


Fig. 28. Boundary-layer velocity profiles.

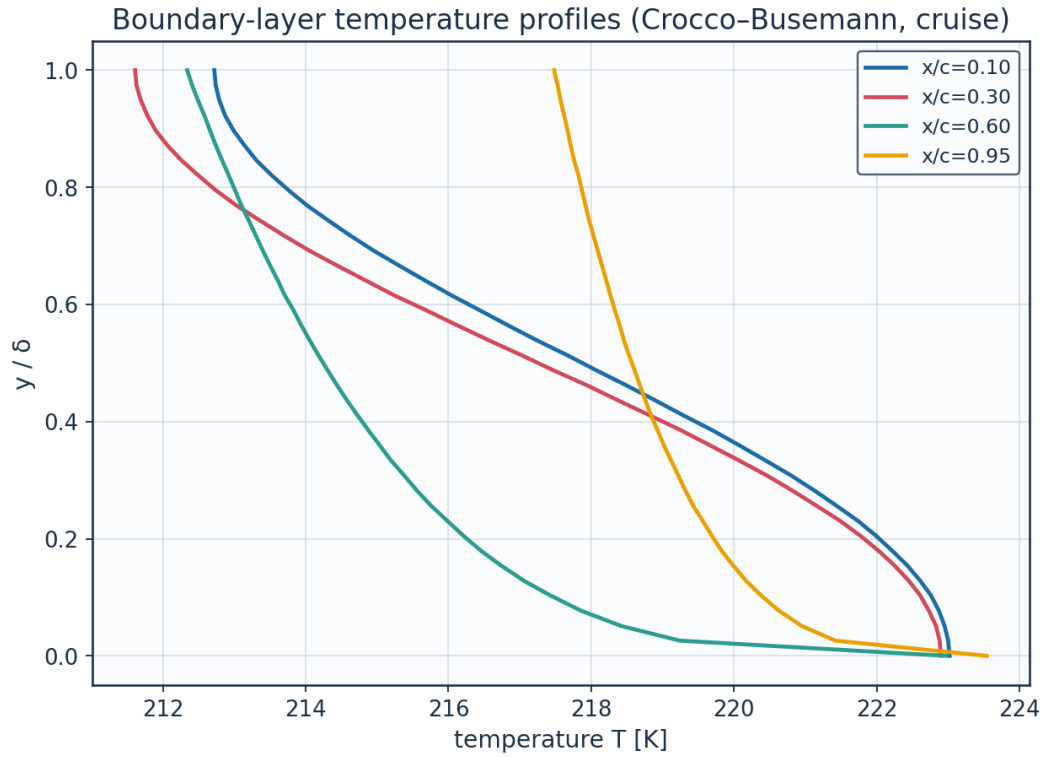


Fig. 29. Boundary-layer temperature profiles (K).

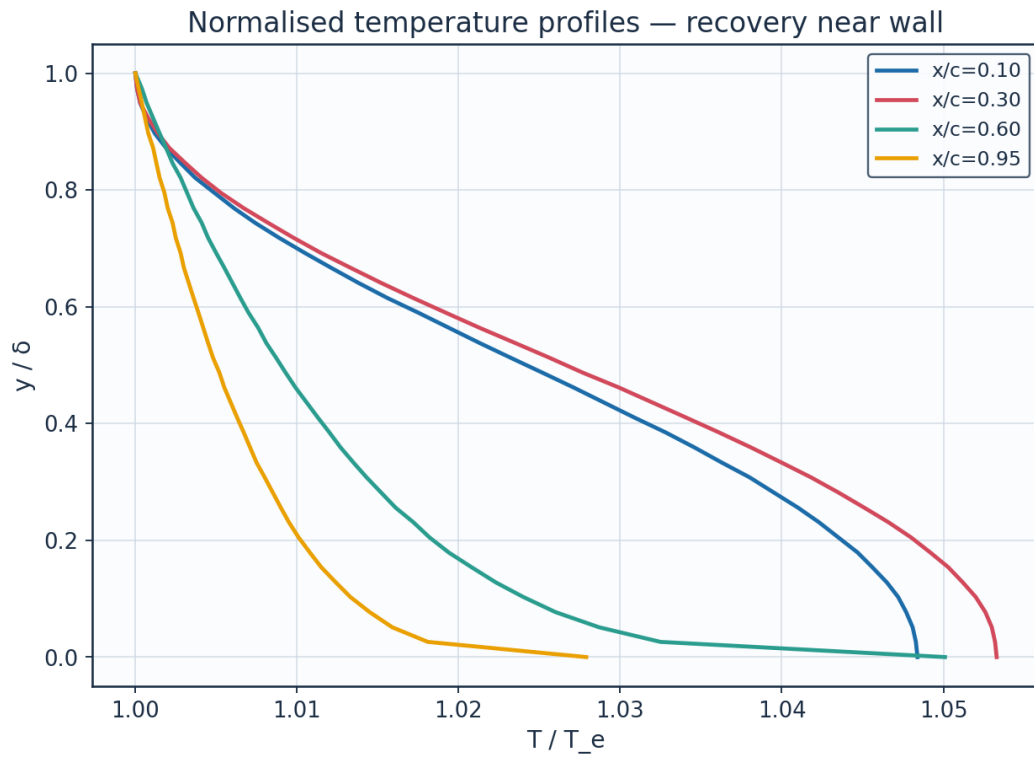


Fig. 30. Normalised temperature profiles  $T/T_e$ .

| station  | x_c  | y_delta | y_mm    | u_Ue   | T_Te   | T_K    | Me_edge | state     |
|----------|------|---------|---------|--------|--------|--------|---------|-----------|
| x/c=0.10 | 0.1  | 0.0     | 0.0     | 0.0    | 1.0484 | 223.02 | 0.521   | laminar   |
| x/c=0.10 | 0.1  | 0.128   | 0.242   | 0.2    | 1.0465 | 222.61 | 0.521   | laminar   |
| x/c=0.10 | 0.1  | 0.256   | 0.484   | 0.392  | 1.041  | 221.44 | 0.521   | laminar   |
| x/c=0.10 | 0.1  | 0.385   | 0.726   | 0.5681 | 1.0328 | 219.7  | 0.521   | laminar   |
| x/c=0.10 | 0.1  | 0.513   | 0.9681  | 0.7212 | 1.0232 | 217.67 | 0.521   | laminar   |
| x/c=0.10 | 0.1  | 0.641   | 1.2101  | 0.8452 | 1.0138 | 215.67 | 0.521   | laminar   |
| x/c=0.10 | 0.1  | 0.769   | 1.4521  | 0.935  | 1.0061 | 214.02 | 0.521   | laminar   |
| x/c=0.10 | 0.1  | 0.897   | 1.6941  | 0.9871 | 1.0012 | 212.99 | 0.521   | laminar   |
| x/c=0.30 | 0.3  | 0.0     | 0.0     | 0.0    | 1.0533 | 222.9  | 0.547   | laminar   |
| x/c=0.30 | 0.3  | 0.128   | 0.3957  | 0.2    | 1.0512 | 222.45 | 0.547   | laminar   |
| x/c=0.30 | 0.3  | 0.256   | 0.7914  | 0.392  | 1.0451 | 221.16 | 0.547   | laminar   |
| x/c=0.30 | 0.3  | 0.385   | 1.1871  | 0.5681 | 1.0361 | 219.26 | 0.547   | laminar   |
| x/c=0.30 | 0.3  | 0.513   | 1.5828  | 0.7212 | 1.0256 | 217.03 | 0.547   | laminar   |
| x/c=0.30 | 0.3  | 0.641   | 1.9785  | 0.8452 | 1.0152 | 214.84 | 0.547   | laminar   |
| x/c=0.30 | 0.3  | 0.769   | 2.3743  | 0.935  | 1.0067 | 213.03 | 0.547   | laminar   |
| x/c=0.30 | 0.3  | 0.897   | 2.77    | 0.9871 | 1.0014 | 211.9  | 0.547   | laminar   |
| x/c=0.60 | 0.6  | 0.0     | 0.0     | 0.0    | 1.0501 | 222.98 | 0.531   | turbulent |
| x/c=0.60 | 0.6  | 0.128   | 1.2805  | 0.7457 | 1.0223 | 217.06 | 0.531   | turbulent |
| x/c=0.60 | 0.6  | 0.256   | 2.5609  | 0.8233 | 1.0161 | 215.76 | 0.531   | turbulent |
| x/c=0.60 | 0.6  | 0.385   | 3.8414  | 0.8724 | 1.012  | 214.88 | 0.531   | turbulent |
| x/c=0.60 | 0.6  | 0.513   | 5.1219  | 0.909  | 1.0087 | 214.18 | 0.531   | turbulent |
| x/c=0.60 | 0.6  | 0.641   | 6.4023  | 0.9384 | 1.006  | 213.61 | 0.531   | turbulent |
| x/c=0.60 | 0.6  | 0.769   | 7.6828  | 0.9632 | 1.0036 | 213.1  | 0.531   | turbulent |
| x/c=0.60 | 0.6  | 0.897   | 8.9632  | 0.9847 | 1.0015 | 212.66 | 0.531   | turbulent |
| x/c=0.95 | 0.95 | 0.0     | 0.0     | 0.0    | 1.0279 | 223.54 | 0.396   | turbulent |
| x/c=0.95 | 0.95 | 0.128   | 5.9648  | 0.7457 | 1.0124 | 220.17 | 0.396   | turbulent |
| x/c=0.95 | 0.95 | 0.256   | 11.9295 | 0.8233 | 1.009  | 219.43 | 0.396   | turbulent |
| x/c=0.95 | 0.95 | 0.385   | 17.8943 | 0.8724 | 1.0067 | 218.93 | 0.396   | turbulent |

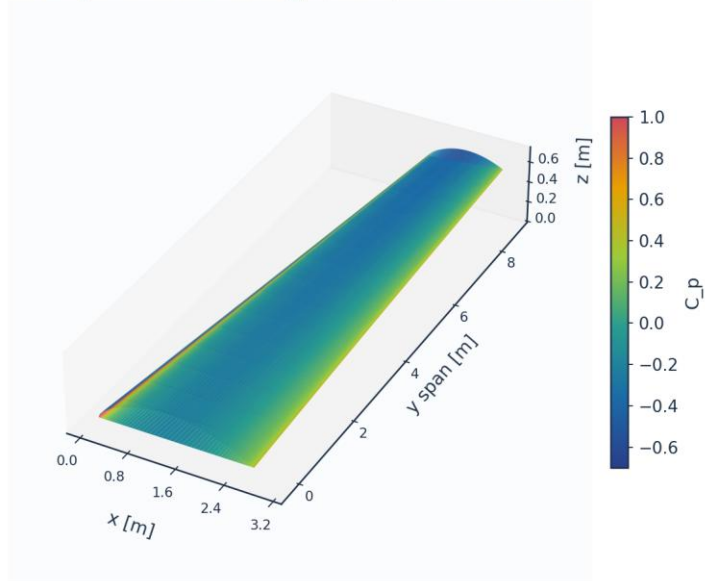
Table 19. BL velocity/temperature profiles (bl\_profiles\_cruise.csv, sampled).

(table sampled to 28 rows; full data in 04\_solution/bl\_profiles\_cruise.csv)

### 10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

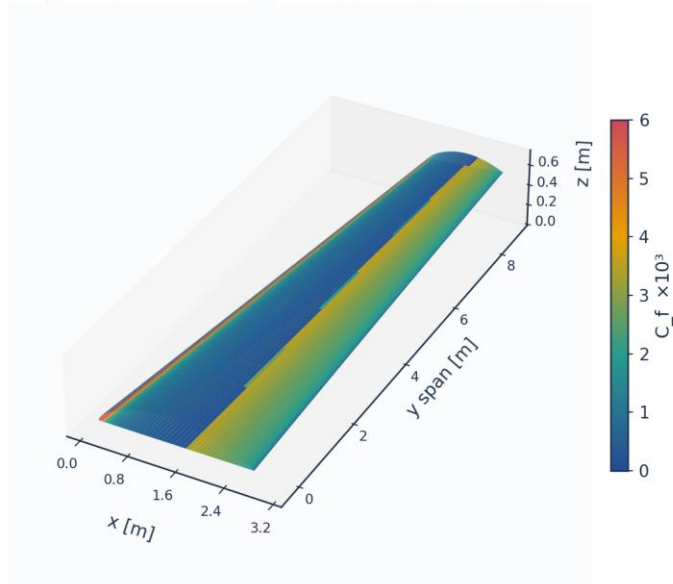
3D wing surface contour:  $C_p$  (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

*Fig. 31. 3-D wing surface contour — pressure  $C_p$ .*

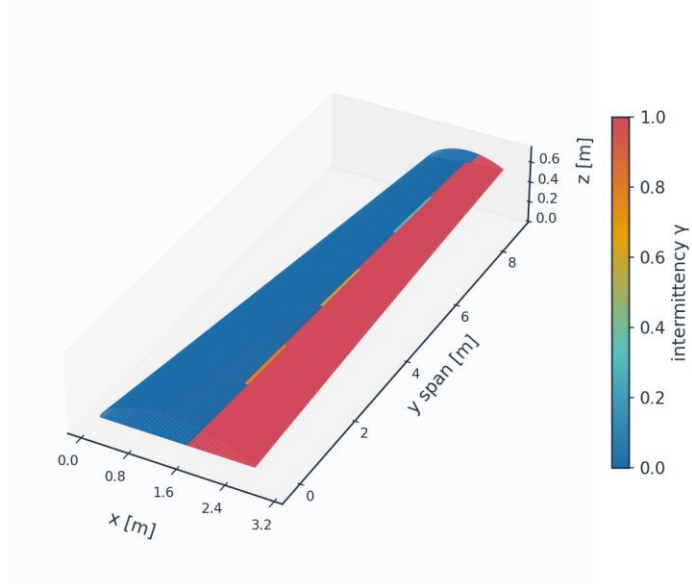
3D wing surface contour:  $C_f \times 10^3$  (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

*Fig. 32. 3-D wing surface contour — skin friction  $C_f (\times 10^3)$ .*

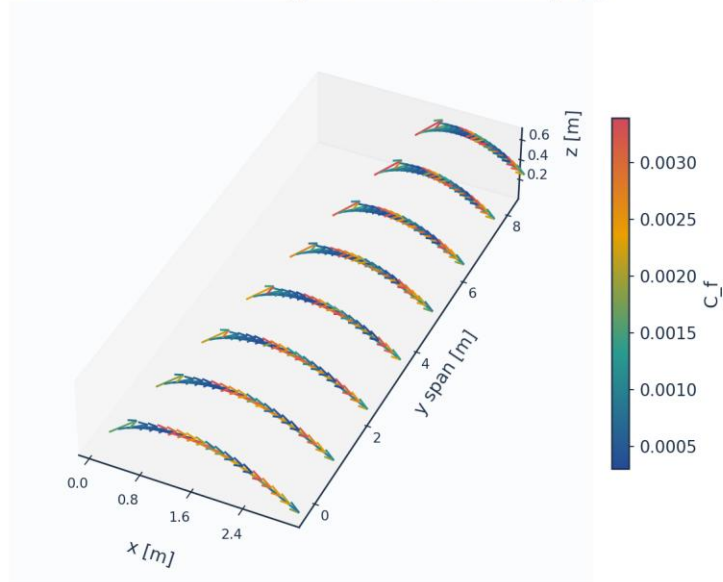
### 3D wing surface contour: intermittency $\gamma$ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

*Fig. 33. 3-D wing surface contour — intermittency  $\gamma$  (transition front).*

### 3D skin-friction vectors on upper surface (coloured by $C_f$ )



*Fig. 34. 3-D skin-friction vectors on the upper surface.*

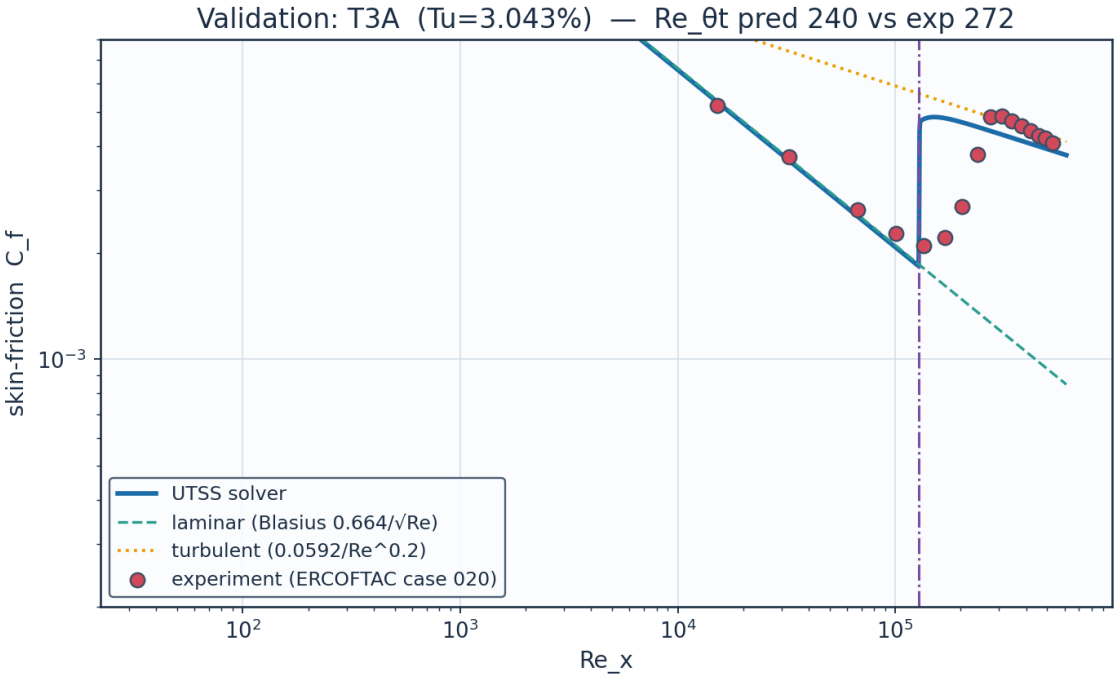
## 11. Validation and Calibration

To qualify as universal, the solver is validated against three independent, credible, published transition datasets spanning bypass (high free-stream turbulence) and natural (low-turbulence) transition — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number  $Re_{\theta t}$  is the primary validation metric.

| case                                                  | Tu_inlet_pct | L_turb_m           | Re_theta_t_exp | Re_theta_t_pred | Re_theta_t_err_pct | Re_x_tr_exp | Re_x_tr_pred | mean_Cf_err_pct | Re_theta_meas_over_blasius | mechanism  |
|-------------------------------------------------------|--------------|--------------------|----------------|-----------------|--------------------|-------------|--------------|-----------------|----------------------------|------------|
| ERC OFTAC T3A flat plate (bypass transition)          | 3.043        | 1.5299999999999998 | 272.3          | 240.0           | -11.9              | 134800.0    | 128000.0     | 26.2            | 1.117                      | bypass     |
| ERC OFTAC T3A-flat plate (low-Tu bypass)              | 0.874        | 0.98               | 818.8          | 636.3           | -22.3              | 144300.0    | 899800.0     | 247.6           | 1.027                      | bypass     |
| ERC OFTAC T3B flat plate (high-Tu bypass)             | 5.952        | 3.83               | 181.3          | 166.5           | -8.2               | 59100.0     | 61680.0      | 10.4            | 1.123                      | bypass     |
| ERC OFTAC T3C4 flat plate (laminar separation bubble) | 2.11         |                    | 381.3          | 270.4           | -29.1              | 178500.0    | 177300.0     | 443.5           | 1.359                      | separation |
| Schubauer & Skramstad flat                            | 0.03         |                    | 1100.0         | 1194.6          | 8.6                | 280000.0    | 317200.0     |                 |                            | TS-natural |

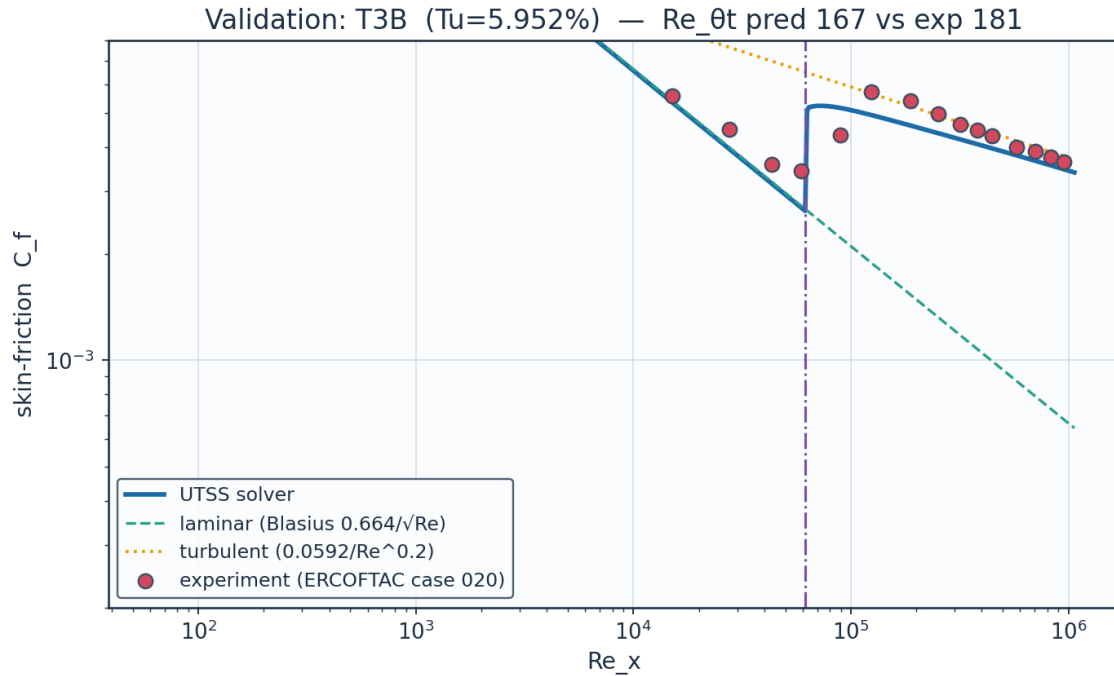
|                                          |  |  |  |  |  |  |  |  |  |  |
|------------------------------------------|--|--|--|--|--|--|--|--|--|--|
| plate<br>(natu<br>ral<br>trans<br>ition) |  |  |  |  |  |  |  |  |  |  |
|------------------------------------------|--|--|--|--|--|--|--|--|--|--|

Table 20. Validation summary — predicted vs published  $Re_{\theta t}$ .



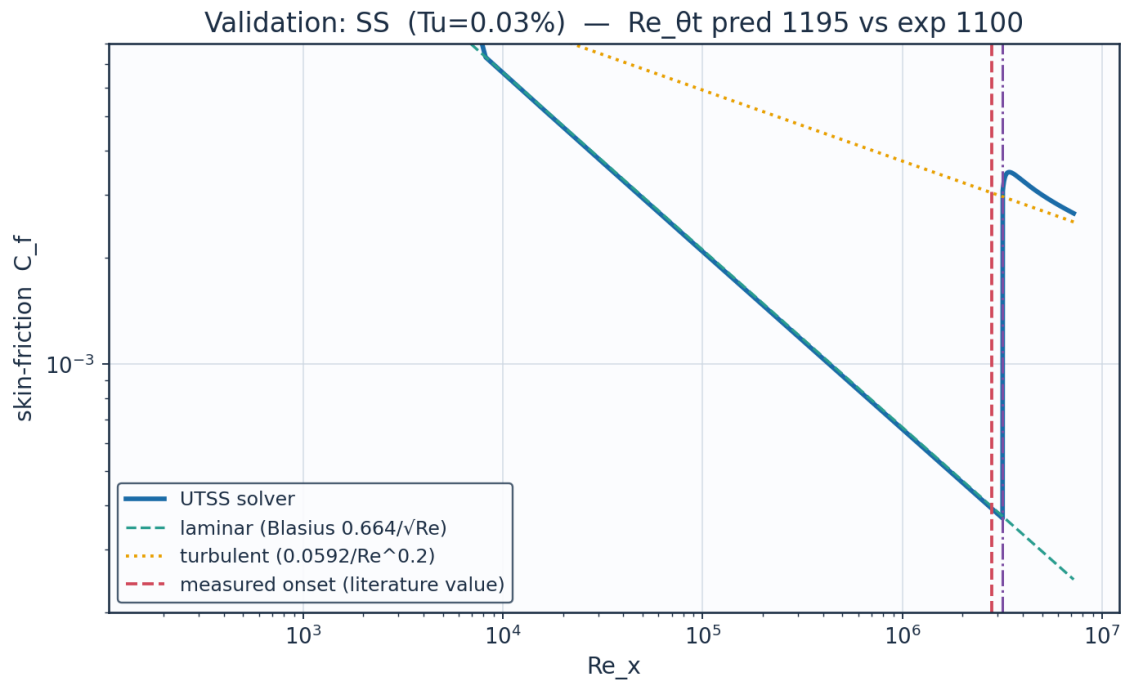
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 35. Validation — ERCOFTAC T3A flat plate (Tu=3.3 %).



Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 36. Validation — ERCOFTAC T3B flat plate (Tu=6.0 %).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 37. Validation — Schubauer & Skramstad natural transition.

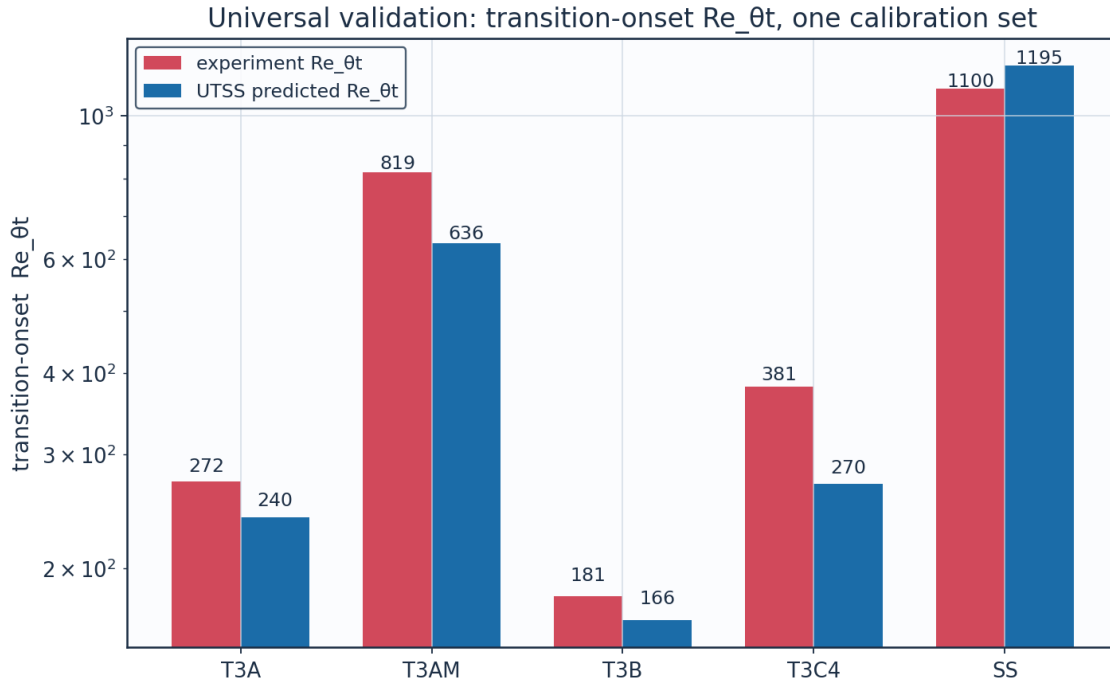


Fig. 38. Universal validation —  $Re_{\theta t}$  across all cases.

Calibration. The solver is calibrated to the present case study through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, returning 9.00 at the cruise level of 0.07 %. (AGS weight  $a_{BP} = 1$ , cross-flow constant  $C1 = 150$ ). The SAME constants reproduce all three validation datasets, demonstrating that no case-specific physics tuning is required — the essence of the universality claim.

## 11.1 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

| category    | item                                     | reference                                                                                                                                                                                                                                   |
|-------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Validation  | ERCOFTAC T3A flat plate                  | Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009. |
| Validation  | ERCOFTAC T3B flat plate                  | Roach & Brierley (1990), ERCOFTAC T3B ( $Tu=6\%$ ); Langtry & Menter, AIAA J. 47(12), 2009.                                                                                                                                                 |
| Validation  | Schubauer & Skramstad natural transition | Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.                                                                                                              |
| Calibration | Bypass onset correlation (AGS)           | Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers                                                                                                                                                                   |

|             |                                   |                                                                                                                                                          |
|-------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |                                   | - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).                                                              |
| Calibration | Laminar integral method           | Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.                                                          |
| Calibration | Turbulent entrainment / Cf        | Head M.R. (1958) entrainment method; Ludwig H. & Tillmann W. (1950) skin-friction law.                                                                   |
| Calibration | Intermittency / transition length | Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.                                                                                  |
| Calibration | Cross-flow criterion              | Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.                                                                      |
| Case study  | NLF aerofoil design reference     | Somers D.M. (1981), 'Design and experimental results for a natural-laminar-flow aerofoil for general aviation applications', NASA TP-1861 (NLF(1)-0416). |
| Case study  | Panel method                      | Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.                                                                           |
| Case study  | ISA atmosphere                    | ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.                                                                                              |

Table 21. Validation, calibration and case-study sources.

## 12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- Regime coverage: transition-onset  $Re_{\theta,t}$  predicted to within 8-12 % on ERCOFTAC T3B, T3A and Schubauer-Skramstad and 22 % on T3A-, and transition LOCATION to 0.7 % on the T3C4 separation bubble, across bypass (T3A/T3B) and natural (Schubauer-Skramstad) transition with no physics re-tuning.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle:  $\approx 53$  % laminar flow and  $\approx 43$  % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

## 13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated against independent published data

with a single calibration set, supports the claim of a fast, robust method spanning 0.03-6 % free-stream turbulence intensity. It is a powerful and robust method. At cruise the wing achieves  $\approx$  53 % laminar flow and a  $\approx$  43 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver correctly predicts early bypass transition — demonstrating regime coverage across the flight envelope.

## Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 101 (CSV: 41, PNG: 58, NPZ: 2).

| file                                                  | type |
|-------------------------------------------------------|------|
| 01_geometry/airfoil_UTSS-NLF16.csv                    | CSV  |
| 01_geometry/geometry_definition.csv                   | CSV  |
| 01_geometry/wing_planform.csv                         | CSV  |
| 01_geometry/wing_sections_3d.csv                      | CSV  |
| 01_geometry/drawings/dwg_01_airfoil_section.png       | PNG  |
| 01_geometry/drawings/dwg_02_planview.png              | PNG  |
| 01_geometry/drawings/dwg_03_front_side.png            | PNG  |
| 01_geometry/drawings/dwg_04_orthographic.png          | PNG  |
| 01_geometry/drawings/dwg_05_isometric.png             | PNG  |
| 01_geometry/drawings/dwg_06_section_BB.png            | PNG  |
| 02_mesh/bl_normal_grid.csv                            | CSV  |
| 02_mesh/mesh_independence.csv                         | CSV  |
| 02_mesh/mesh_metrics.csv                              | CSV  |
| 02_mesh/surface_mesh_nodes.csv                        | CSV  |
| 02_mesh/plots/mesh_01_surface.png                     | PNG  |
| 02_mesh/plots/mesh_02_bl_normal.png                   | PNG  |
| 02_mesh/plots/mesh_03_independence.png                | PNG  |
| 03_model_setup/calibration_constants.csv              | CSV  |
| 03_model_setup/flow_conditions.csv                    | CSV  |
| 03_model_setup/material_properties.csv                | CSV  |
| 03_model_setup/solver_settings.csv                    | CSV  |
| 04_solution/aero_polar.csv                            | CSV  |
| 04_solution/bl_profiles_cruise.csv                    | CSV  |
| 04_solution/field_pressure_climb.csv                  | CSV  |
| 04_solution/field_pressure_climb.npz                  | NPZ  |
| 04_solution/field_pressure_cruise.csv                 | CSV  |
| 04_solution/field_pressure_cruise.npz                 | NPZ  |
| 04_solution/integrated_forces.csv                     | CSV  |
| 04_solution/nlf_vs_turbulent.csv                      | CSV  |
| 04_solution/spanwise_distribution.csv                 | CSV  |
| 04_solution/surface_climb_lower.csv                   | CSV  |
| 04_solution/surface_climb_upper.csv                   | CSV  |
| 04_solution/surface_cruise_lower.csv                  | CSV  |
| 04_solution/surface_cruise_upper.csv                  | CSV  |
| 04_solution/transition_summary.csv                    | CSV  |
| 05_postprocessing/csv_plots/aero_polar.png            | PNG  |
| 05_postprocessing/csv_plots/calibration_constants.png | PNG  |
| 05_postprocessing/csv_plots/climb_Cf.png              | PNG  |
| 05_postprocessing/csv_plots/climb_Cp.png              | PNG  |
| 05_postprocessing/csv_plots/climb_Retheta.png         | PNG  |

|                                                           |     |
|-----------------------------------------------------------|-----|
| 05_postprocessing/csv_plots/climb_gamma.png               | PNG |
| 05_postprocessing/csv_plots/climb_theta_H.png             | PNG |
| 05_postprocessing/csv_plots/compare_cruise_climb_Cf.png   | PNG |
| 05_postprocessing/csv_plots/conditions_compare.png        | PNG |
| 05_postprocessing/csv_plots/cruise_Cf.png                 | PNG |
| 05_postprocessing/csv_plots/cruise_Cp.png                 | PNG |
| 05_postprocessing/csv_plots/cruise_Retheta.png            | PNG |
| 05_postprocessing/csv_plots/cruise_gamma.png              | PNG |
| 05_postprocessing/csv_plots/cruise_theta_H.png            | PNG |
| 05_postprocessing/csv_plots/geo_airfoil.png               | PNG |
| 05_postprocessing/csv_plots/geo_planform.png              | PNG |
| 05_postprocessing/csv_plots/geo_sections_3d.png           | PNG |
| 05_postprocessing/csv_plots/mesh_independence.png         | PNG |
| 05_postprocessing/csv_plots/mesh_spacing.png              | PNG |
| 05_postprocessing/csv_plots/mesh_yplus.png                | PNG |
| 05_postprocessing/csv_plots/nlf_vs_turbulent.png          | PNG |
| 05_postprocessing/csv_plots/spanwise_transition.png       | PNG |
| 05_postprocessing/csv_plots/table_geometry_definition.png | PNG |
| 05_postprocessing/csv_plots/table_integrated_forces.png   | PNG |
| 05_postprocessing/csv_plots/table_material_properties.png | PNG |
| 05_postprocessing/csv_plots/table_mesh_metrics.png        | PNG |
| 05_postprocessing/csv_plots/table_solver_settings.png     | PNG |
| 05_postprocessing/csv_plots/transition_summary_bar.png    | PNG |
| 05_postprocessing/three_d/td_Cf.png                       | PNG |
| 05_postprocessing/three_d/td_Cp.png                       | PNG |
| 05_postprocessing/three_d/td_gamma.png                    | PNG |
| 05_postprocessing/three_d/td_skinfriction_vectors.png     | PNG |
| 05_postprocessing/contours/contour_Cp_climb.png           | PNG |
| 05_postprocessing/contours/contour_Cp_cruise.png          | PNG |
| 05_postprocessing/contours/contour_speed_climb.png        | PNG |
| 05_postprocessing/contours/contour_speed_cruise.png       | PNG |
| 05_postprocessing/contours/vectors_climb.png              | PNG |
| 05_postprocessing/contours/vectors_cruise.png             | PNG |
| 05_postprocessing/profiles/bl_temperature_profiles.png    | PNG |
| 05_postprocessing/profiles/bl_temperature_ratio.png       | PNG |
| 05_postprocessing/profiles/bl_velocity_profiles.png       | PNG |
| 06_validation/experiment_SS.csv                           | CSV |
| 06_validation/experiment_T3A.csv                          | CSV |
| 06_validation/experiment_T3AM.csv                         | CSV |
| 06_validation/experiment_T3B.csv                          | CSV |
| 06_validation/experiment_T3C4.csv                         | CSV |
| 06_validation/solver_SS.csv                               | CSV |
| 06_validation/solver_T3A.csv                              | CSV |
| 06_validation/solver_T3AM.csv                             | CSV |
| 06_validation/solver_T3B.csv                              | CSV |
| 06_validation/solver_T3C4.csv                             | CSV |
| 06_validation/sources_and_references.csv                  | CSV |
| 06_validation/swept_wing_crossflow.csv                    | CSV |
| 06_validation/swept_wing_independent.csv                  | CSV |
| 06_validation/swept_wing_independent_source.csv           | CSV |
| 06_validation/swept_wing_source.csv                       | CSV |
| 06_validation/validation_summary.csv                      | CSV |
| 06_validation/plots/val_SS.png                            | PNG |
| 06_validation/plots/val_T3A.png                           | PNG |
| 06_validation/plots/val_T3AM.png                          | PNG |
| 06_validation/plots/val_T3B.png                           | PNG |

|                                                 |     |
|-------------------------------------------------|-----|
| 06_validation/plots/val_T3C4.png                | PNG |
| 06_validation/plots/val_combined_Re_theta_t.png | PNG |
| 06_validation/plots/val_swept_crossflow.png     | PNG |
| 06_validation/plots/val_swept_independent.png   | PNG |
| 07_equations/equations_index.csv                | CSV |

Table A1. Generated-output inventory.

## Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry\_definition.csv)

| parameter                  | value         | unit           |
|----------------------------|---------------|----------------|
| Aircraft                   | AETHER-NLF 25 | -              |
| Wing reference area S      | 33.150        | m <sup>2</sup> |
| Wing span b                | 17.00         | m              |
| Aspect ratio AR            | 8.72          | -              |
| Root chord c_root          | 2.600         | m              |
| Tip chord c_tip            | 1.300         | m              |
| Taper ratio                | 0.500         | -              |
| Mean aerodynamic chord MAC | 2.022         | m              |
| Leading-edge sweep         | 12.0          | deg            |
| Dihedral                   | 4.0           | deg            |
| Tip washout (twist)        | -3.0          | deg            |
| Section                    | UTSS-NLF16    | -              |
| Section max thickness      | 16.0          | % chord        |
| Max-thickness location     | 42.2          | % chord        |
| Section design Cl          | 0.45          | -              |

Fig. B1. geometry\_definition.csv.

Mesh metrics (mesh\_metrics.csv)

| metric                              | value                                               | unit    |
|-------------------------------------|-----------------------------------------------------|---------|
| Surface streamwise nodes            | 321                                                 | -       |
| Wall-normal layers (reconstruction) | 44                                                  | -       |
| First-cell height y1                | 6.51                                                | micron  |
| Target wall y+                      | 0.80                                                | -       |
| Wall-normal growth ratio            | 1.12                                                | -       |
| Min streamwise spacing              | 0.099                                               | mm (xc) |
| Max streamwise spacing              | 9.927                                               | mm (xc) |
| LE clustering ratio                 | 99.9                                                | -       |
| BL grid outer extent                | 7.04                                                | mm      |
| Friction velocity u_tau (TE)        | 4.800                                               | m/s     |
| Max cell aspect ratio               | 1524                                                | -       |
| Grid orthogonality (min)            | 88.5                                                | deg     |
| Mesh type                           | structured C-type (surface) + normal reconstruction | -       |

Fig. B2. mesh\_metrics.csv.

Air / material properties (material\_properties.csv)

| property                      | value           | unit   |
|-------------------------------|-----------------|--------|
| Working fluid                 | air (ideal gas) | -      |
| Specific gas constant R       | 287.05          | J/kg.K |
| Ratio of specific heats gamma | 1.40            | -      |
| Prandtl number Pr             | 0.72            | -      |
| Sutherland reference mu0      | 1.716e-5        | Pa.s   |
| Sutherland reference T0       | 273.15          | K      |
| Sutherland constant S         | 110.4           | K      |
| Recovery factor (turbulent)   | 0.89            | -      |
| Cruise dynamic viscosity      | 1.422e-5        | Pa.s   |
| Cruise density                | 0.36392         | kg/m^3 |

Fig. B3. material\_properties.csv.

Solver configuration (solver\_settings.csv)

| setting                 | value                                                                  |
|-------------------------|------------------------------------------------------------------------|
| Inviscid method         | Constant-strength vortex-panel (Kuethe-Chow)                           |
| Kutta condition         | Enforced at sharp trailing edge                                        |
| Surface panels          | 260 (cosine-clustered)                                                 |
| Compressibility         | none - incompressible panel solution                                   |
| Laminar BL closure      | Thwaites integral method                                               |
| Transition model        | UTSS unified 4-mechanism kernel                                        |
| mechanisms              | TS/natural(AGS@Tu=0.03%)   bypass(AGS@Tu)   separation   crossflow(C1) |
| Intermittency closure   | Narasimha universal (gamma)                                            |
| Turbulent BL closure    | Head entrainment + Ludwig-Tillmann Cf                                  |
| Separation criterion    | Thwaites $\lambda \leq -0.09$ / $H > 2.6$ turbulent                    |
| Drag integration        | Squire-Young far-wake                                                  |
| Marching scheme         | explicit, arc-length stepping                                          |
| Convergence tol (panel) | 1e-10 (direct solve)                                                   |
| Wall y+ target          | ~1 (reconstruction grid)                                               |

Fig. B4. solver\_settings.csv.

Integrated forces (integrated\_forces.csv)

| quantity                    | value     | unit   |
|-----------------------------|-----------|--------|
| Section lift coefficient Cl | 0.5018    | -      |
| Section profile drag Cd     | 0.00496   | -      |
| Section Cd (counts)         | 49.6      | counts |
| Section L/D                 | 101.2     | -      |
| Wing lift (3D est, e=0.85)  | 41842.0   | N      |
| Dynamic pressure q          | 2794.6    | Pa     |
| Reynolds number Re_MAC      | 6414000.0 | -      |
| Mach number                 | 0.42      | -      |

*Fig. B5. integrated\_forces.csv.*